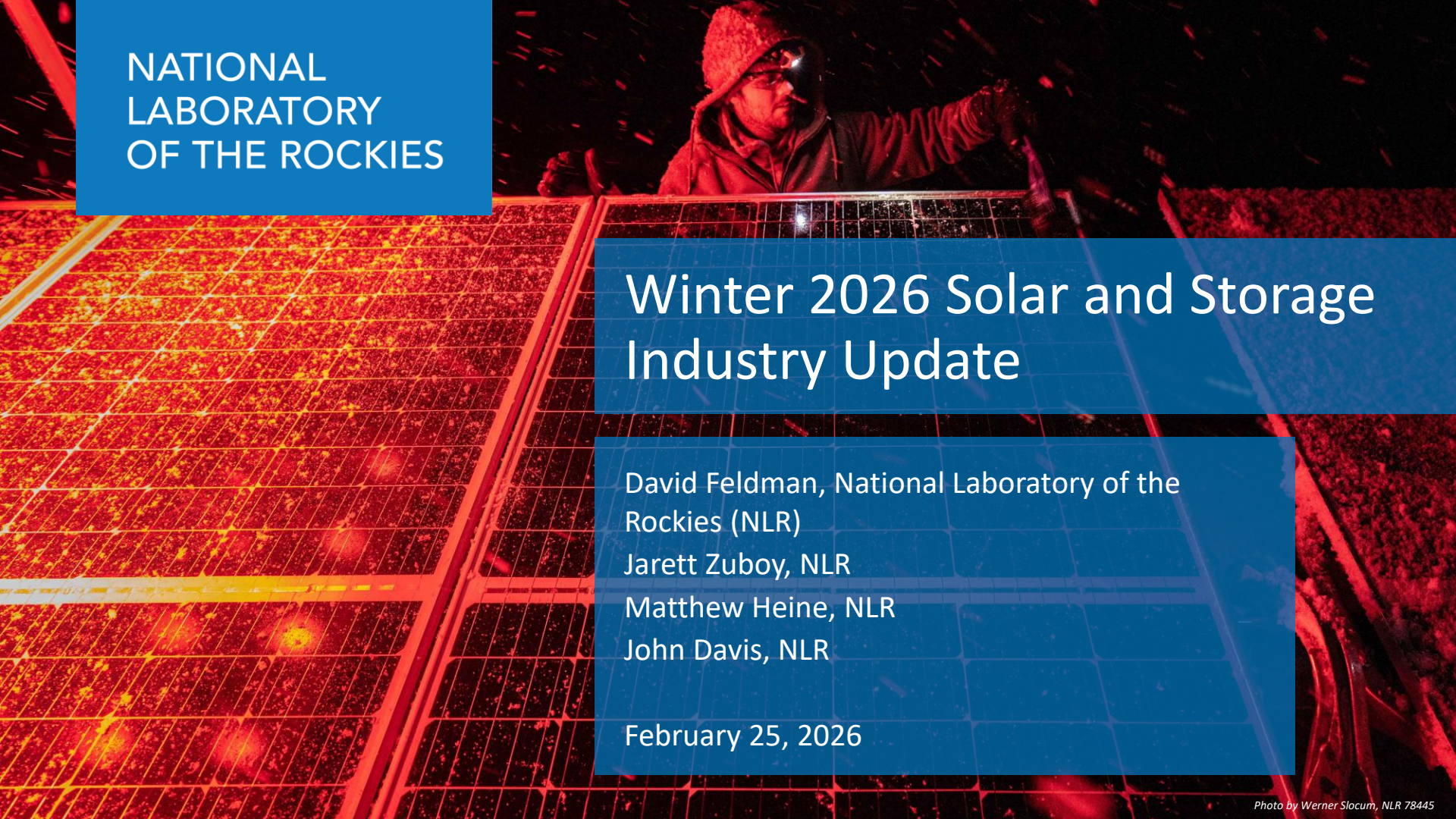


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A full-page background image showing a worker in a winter hat and safety glasses cleaning large solar panels. The scene is covered in snow, and the worker is using a tool to clear the panels. The image has a blue overlay on the right side where the text is located.

NATIONAL
LABORATORY
OF THE ROCKIES

Winter 2026 Solar and Storage Industry Update

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February 25, 2026

Agenda

- 1 Global Solar Deployment**
- 2 U.S. PV and BESS Deployment**
- 3 PV System and BESS Pricing**
- 4 Global Manufacturing**
- 5 Component Pricing**
- 6 Market and Policy**
- 7 U.S. PV Imports**

Executive Summary

Global Solar Deployment

- Analysts estimate that about 655 GW_{dc} of PV were installed globally in 2025, up 18% from 2024.
- China installed 311 GW_{dc} of PV in 2025 to surpass 1 TW_{dc} cumulatively; however, Q4 installs were down 28%, y/y.
- India's 2025 installations were up 64% to around 38 GW_{dc} as long-delayed projects were commissioned to avoid transmission charges; rooftop installations made up about 20% of the total, and off-grid applications made up over 1 GW.

U.S. PV and BESS Deployment

- In 2025, the United States installed approximately 34.0 GW_{ac} of PV (SEIA reported 43.2 GW_{dc}), 15.2 GW_{ac} (41.9 GWh) of grid-scale storage, and 5.9 GW_{ac} of land-based wind.
- PV installations were down 8% by capacity, y/y, with fewer large-scale projects.
- About 2.1 GW_{ac} of distributed BESS were installed.
- EIA projects that U.S. PV's annual growth will rise to 38.1 GW_{ac} in 2026 and 43.4 GW_{ac} in 2027. Other analyst estimates project 37 GW_{dc} to 44 GW_{dc} of annual PV deployment through 2029.

PV System and BESS Pricing

- LevelTen reports that, in Q4 2025, U.S. utility-scale photovoltaic PPA prices increased for the fifth quarter in a row, up 9%, y/y.
- From H2 2024 to H2 2025, the median real reported distributed PV system price (no storage) across AZ, CA, MA, and NY fell for smaller systems and generally rose for larger systems.
- Recent increases in Sunrun installation costs were due in part to increasing battery attachment rates, which grew from 27% in 2023 to 70% in 2025.

Market and Policy

- Regulators in Nevada, Virginia, and West Virginia reached decisions on net metering successor tariff proposals.
- As of October 2025, at least 10 states have implemented policies in direct response to declining federal support.
- At the end of 2024, more than 280,000 U.S. employees spent most of their time on solar project- or manufacturing-related work, flat, y/y. An additional 90,000 workers spent less than half their time on solar-related work. Another 79,000 jobs were in BESS in 2024, a growth of 4%, y/y.

Executive Summary (Continued)

Global Manufacturing

- Margins for the leading PV wafer, cell, and module manufacturers continued to recover in Q3 2025, though most companies were still losing money.
- BloombergNEF reported that, at the end of 2025, the approximate global c-Si PV manufacturing capacity was between 1.35 and 1.67 TW—a growth of 2% (c-Si wafers) to 10% (c-Si modules)—with most still in China.
 - Due to manufacturing capacity built in 2025, at the end of the year, 40% of c-Si manufacturing capacity outside of China was in India.
- The United States manufactured approximately 24.9 GW_{dc} of PV panels in the first 9 months of 2025, up 103%, y/y, though flat, q/q.
- At the end of Q3 2025, the United States had a c-Si cell manufacturing capacity of 3 GW_{dc}.

Component Pricing

- Global polysilicon, wafer, and cell pricing increased 30%–44% in 2025, attributed to Chinese government intervention and the rising price of silver; global module prices were around 0.10/W_{dc} at the end of January 2026.

- The U.S. solar supply chain, however, is only indirectly impacted by Chinese market conditions due to tariffs and other trade restrictions; module prices in the United States remained around \$0.27/W_{dc} at the end of January 2026.
- The import value of PV modules decreased dramatically in H2 2025; however, tariffs are likely keeping the price-to-consumer at relatively steady levels.
- After a small increase in 2022, stationary storage battery pack prices continued their historical downward trend in 2025, falling 84% from 2016 to 2025 and 45% between 2024 and 2025, to \$70/kWh.

U.S. PV Imports

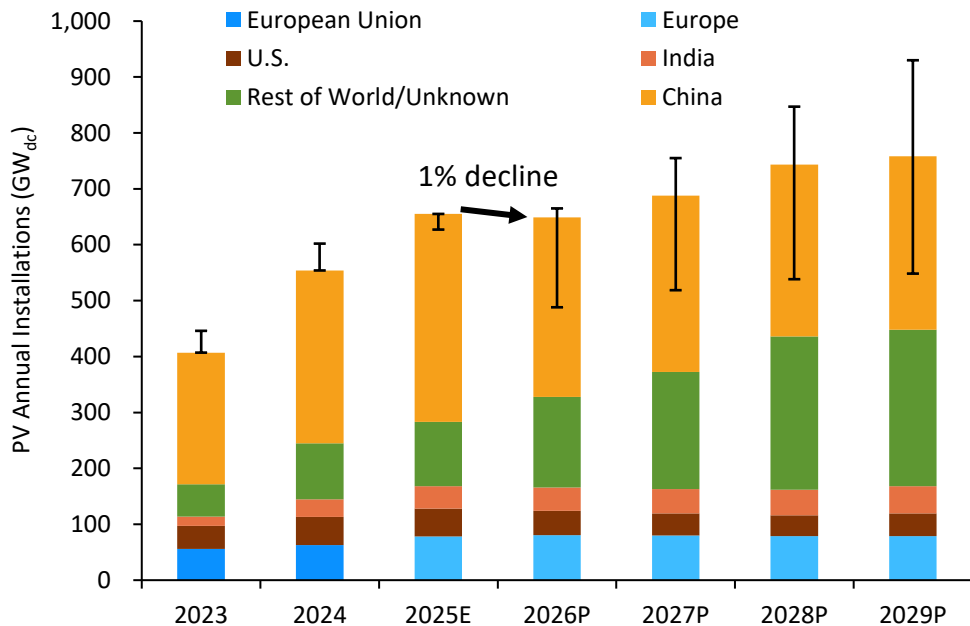
- The United States imported 33.2 GW_{dc} of PV modules in 2025, a 40% decrease from 2024.
- The United States imported 21.1 GW_{dc} of PV cells in 2025, a 50% increase from 2024.
- Indonesia and Laos represented about half the imports of c-Si module and cell imports in 2025.

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- India's 2025 installations were up 64% to around 38 GW_{dc} as long-delayed projects were commissioned to avoid transmission charges.
 - Rooftop installations made up about 20% of the total, and off-grid applications made up over 1 GW_{dc}.

Annual Global PV Deployment



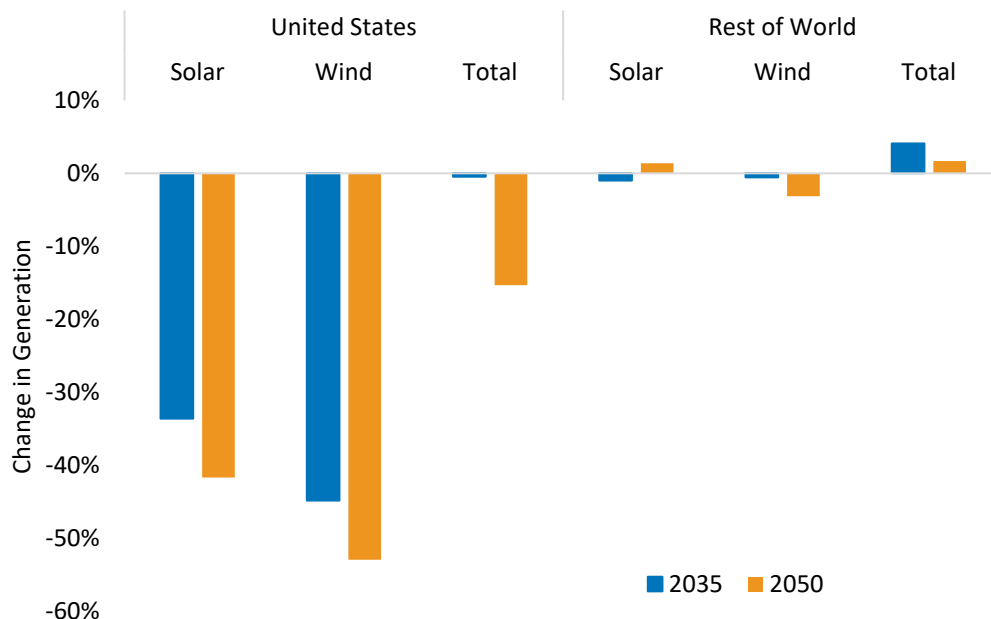
- The median of estimates suggests about 655 GW_{dc} of PV were installed globally in 2025, up 18% from 2024.
- Projections suggest lower global growth than the historical average, as annual installations decline 1% in 2026 and average only 3% growth from 2025 to 2029, reaching 758 GW_{dc}.
 - BNEF reports the decline expected in 2026 will be the first in more than 2 decades of tracking.
 - BNEF attributes the slowing to policy changes in China and the United States as well as saturation in some mature markets such as Brazil and Spain.
- Among analysts covered in the Winter 2025 *Solar Industry Update*, changes in projections in this edition ranged from -9% in 2026 to +3% in 2027 and 2028.

	Total Share, 2025–2029	Change in Annual Installs, 2025–2029
China	47%	-16%
Europe	11%	+1%
U.S.	6%	-18%
India	6%	+21%

Notes: E = estimated; P = projected. Bar totals represent median global projections across analysts who provide a global projection. Error bars represent high and low global projections. Regional bar segments represent medians of all available regional projections. Where regional medians do not sum to global medians, the differences are reconciled by adjusting the Rest of World/Unknown segments so that the correct global median values are retained. **Sources:** BloombergNEF (BNEF), 4Q 2025 Global PV Market Outlook, Dec. 2025; EIA, [Annual Energy Outlook 2025](#), April 2025; IEA, [Snapshot of Global PV Markets](#), April 2024 and April 2025; SolarPower Europe, [Global Market Outlook for Solar Power 2025–2029](#), May 2025; Wood Mackenzie, Q4 2025 Solar Executive Briefing, Oct. 2025.

World Energy Outlook 2025

Changes in Electricity Generation from WEO 2024 to WEO 2025 (Stated Policies Scenario)



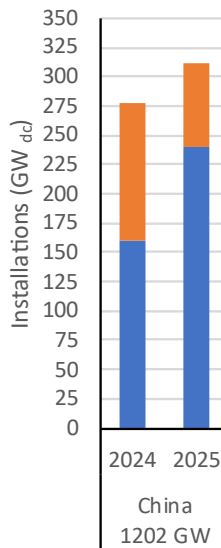
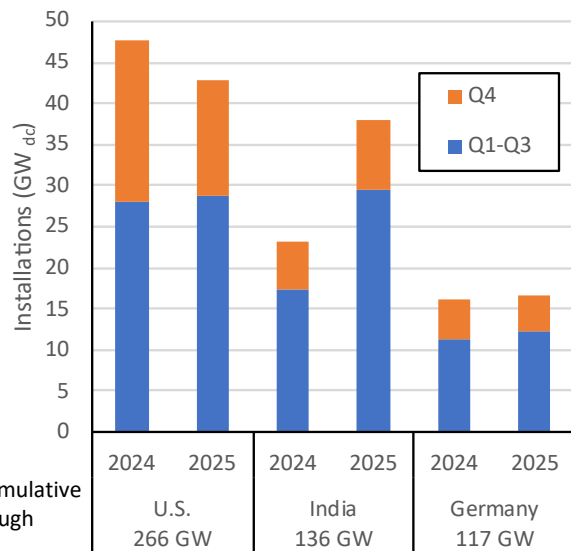
Compared with the International Energy Agency's 2024 World Energy Outlook (WEO), U.S. solar and wind generation decline 34% and 45% by 2035 (42% and 53% by 2050) in the 2025 WEO.

- This comparative decline is seen in the WEO Stated Policies Scenario (STEPS), an exploratory scenario based on national energy policies in place or planned—which is not a forecast.
- In the 2025 WEO STEPS, U.S. solar and wind generation still increase 240% and 60% by 2035 (620% and 100% by 2050) vs. the 2024 baseline.
- U.S. policies emerging between the 2024 WEO and the 2025 WEO resulted in lower solar and wind generation in the 2025 WEO STEPS.

In contrast, future solar and wind generation in STEPS for the rest of the world (ROW) are similar in the 2024 WEO and the 2025 WEO.

- In the 2025 WEO STEPS, ROW solar and wind generation increase 390% and 170% by 2035 (880% and 330% by 2050) vs. the 2024 baseline.
- In the 2025 WEO STEPS, ROW accounts for 85% of solar and 80% of wind generation in 2024, rising to 90% for both resources in 2035 and 2050.

PV Installed in Major Global Markets: 2024 and 2025



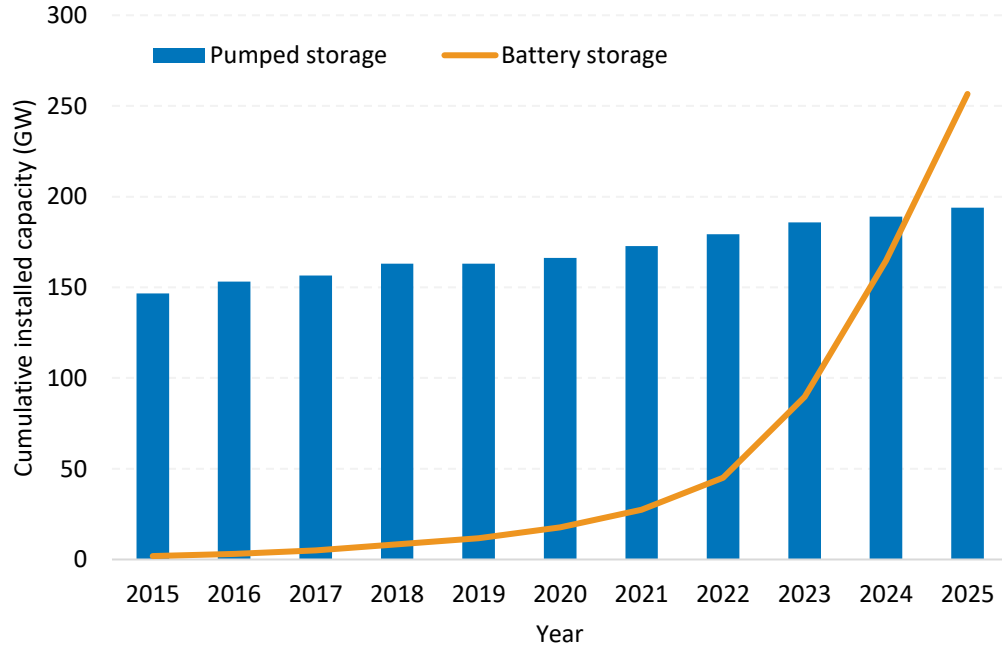
- In 2025, PV installed capacity increased (y/y) in China (+12%) and India (+64%), stayed relatively flat in Germany (+2%), and shrank in the United States (−10%).
 - Q4 2025 installations were down, y/y, in China (−36%), the United States (−28%), and Germany (−11%) but were up 49% in India.
- At the end of 2025, non-fossil sources accounted for 60% of Chinese installed capacity. [PV Magazine](#) reported that with Chinese solar capacity above 1 TW, the market is more focused on system integration than just growth.
- [PV Magazine](#) reported that a large reason for India's y/y solar market growth was due to the commissioning of long-delayed projects to meet a June installation deadline for receiving a waiver for inter-state transmission charges.
 - 2025 Indian solar installations included 28.6 GW_{dc} of utility-scale PV, 7.9 GW_{dc} of rooftop PV, and 1.4 GW_{dc} of off-grid and other distributed PV.

Note: U.S. data converted to DC using a 1.30 and 1.21 AC-to-DC ratio for utility-scale and distributed PV, respectively.

Sources: [Bundesnetzagentur - Erneuerbare Energien](#); EIA, [Electric Power Monthly](#), Nov. 2025; India Ministry of New and Renewable Energy, [Physical Achievements](#), n.d.; JMK Research, [Monthly RE Update – October 2025](#), Oct. 2025; [Monthly RE Update – October 2024](#), October, 2024; Mercom India, [China Adds 28 GW of Solar Capacity in Q3 2025, Down 52% YoY](#), Oct. 2025; PV Magazine, [India Installs Record 29.5 GW of Solar in First Nine Months of 2025](#), Oct. 2025; PV Magazine, [India Deploys 37.9 GW of Solar in 2025](#), Jan. 2026; PV Magazine, [China Adds 315 GW of Solar in 2025](#), Jan. 2026.

Global Battery Capacity Surpasses Pumped Hydro

Cumulative global power capacity: pumped hydro vs. batteries



The global cumulative **power** capacity of batteries for electric-sector energy storage surpassed the capacity of pumped hydro storage for the first time in 2025.

- Battery capacity reached ~260 GW in 2025, while pumped hydro reached ~190 GW.
- Annual growth during 2015–2025 averaged 63% for batteries vs. 3% for pumped hydro.

Pumped hydro provides longer storage **duration** compared with today's battery technologies.

- Duration is the amount of time a storage technology can discharge at its rated power capacity.
- Pumped hydro durations are typically 5–175 hours vs. typical battery durations of 4 hours or less.
- In 2025, average duration for pumped hydro was ~60 hours vs. ~2 hours for batteries.

China hosts 30% of the world's pumped hydro power capacity and 50% of the battery power capacity.

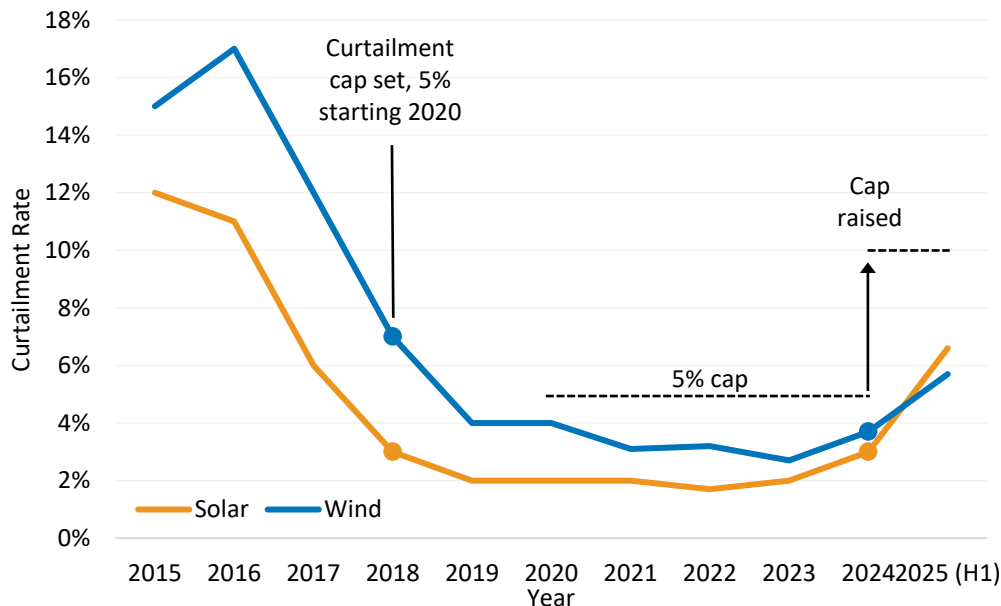
- Other pumped hydro leaders: Europe (30%), Japan (15%), the United States (12%).
- Other battery leaders: Europe (19%), the U.S. (17%).

Projections suggest global pumped hydro installations could reach ~280 GW by 2030 vs. ~990 GW for batteries.

Sources: BNEF, 4Q 2025 Global PV Market Outlook, Dec. 2025; IEA, [How Rapidly Will the Global Electricity Storage Market Grow by 2026?](#), Dec. 2021; International Hydropower Association, [2025 World Hydropower Outlook](#), 2025; PV Magazine, [Global BESS Capacity Tops 250 GW, Overtaking Pumped Hydro for First Time](#), Feb. 6, 2026.

Rising Curtailment in China

Solar and Wind Curtailment Rates in China, 2015 – H1 2025



Between H1 2024 and H1 2025, curtailment—the reduction of generation owing to grid constraints or inadequate demand—rose from 3.9% to 6.6% for solar and from 3% to 5.7% for wind in China.

- In 2024, China raised its curtailment cap from 5% to 10% to promote wind and solar deployment.

Some curtailment is inevitable due to the variability of solar and wind generation, and it has pros and cons.

- Overbuilding solar and wind and using curtailment to manage supply-demand mismatches can be more economical than trying to eliminate curtailment.
- However, curtailment reduces efficiency and may discourage investment; BNEF says raising the cap from 5% to 10% can reduce equity internal rate of return by 1.1–1.5 percentage points in northwest China.

Strategies to mitigate curtailment in China include transmission investments, virtual power plants, green power direct connections, and requiring large energy users to consume more solar and wind electricity.

- Still, analysts expect curtailment to continue rising as China continues to pursue its renewable energy goals.

Sources: BNEF, China Builds World's Largest Supergrid for Renewables Push, Dec. 16, 2025; BNEF, China Eases Grip to Boost Renewables But Spurs Profit Risk, June 4, 2024; BNEF, China Hits RPS Target Earlier But Curtailment Worsens, Jan. 27, 2025; BNEF, China Market Outlook 2H 2025, Nov. 25; Reuters, [As China's Renewable Capacity Soars, Utilisation Lags, Data Show](#), Aug. 5, 2025; Reuters, [China's Renewable Curtailment To Present Revenue Risks in Next Decade, Wood Mackenzie Says](#), Nov. 3, 2025.

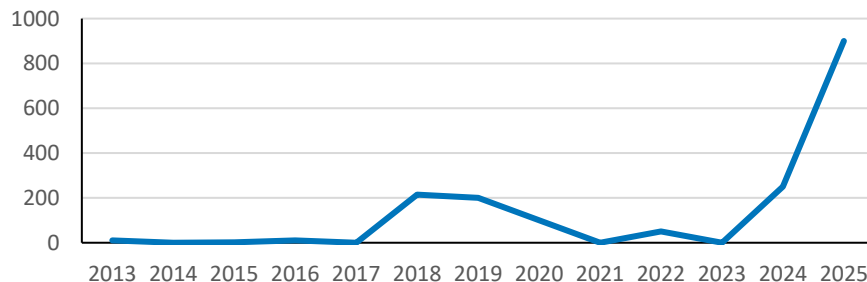
Concentrating Solar Power (CSP) Update

- The European company Volateq has [deployed drones to perform O&M-based inspections](#) for a 100-MW_{ac} parabolic trough CSP plant. The ball joints of parabolic trough collectors require periodic inspection, and the company reported that the automated drones can do this twice as fast as manual inspections.
- Fraunhofer Chile researchers have designed, built, and tested a [Fresnel mirror solar collector and crucible system that melts and recycles aluminum](#). Chile has some of the highest direct normal irradiance, and currently it exports most of its aluminum for recycling. The next steps include refining the design, lowering the costs, and increasing output.
- In late 2025, researchers in France published a new paper [presenting an algorithm strategy to deliver a steady supply of industrial heat](#) from CSP plants with thermal energy storage. The intermittent nature of solar energy can be a barrier to using CSP technology for high-temperature industrial processes with specific temporal requirements.

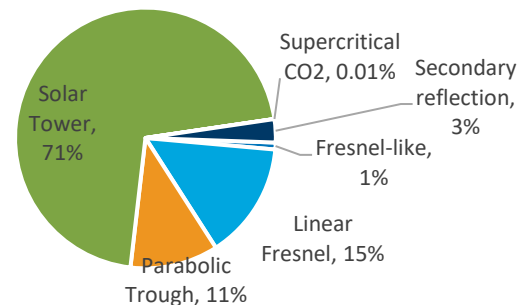
CSP Update (China)

- In 2025, [China connected 9 CSP projects totaling 0.9 GW_{ac}](#) to the electrical grid, bringing the country's cumulative installed CSP capacity to 1.7 GW_{ac}.
 - An additional 25 projects totaling 3 GW_{ac} were also under construction, and another 31 projects, totaling 4 GW_{ac} were planned.
- In December 2025, [China released a policy roadmap](#) that targets 15 GW_{ac} of CSP by 2030 with costs comparable to coal-fired plants (lowering costs from \$72/MWh to \$40/MWh). In addition to R&D and other efforts to lower CSP costs, China plans to increase CSP competitiveness through larger projects, participation in enhancing grid-reliability programs, improved project planning and layout, and integrating projects with PV and wind.
 - To achieve the 15-GW_{ac} goal, the country would need to plan another 6 GW_{ac} of projects.
 - [BNEF reported](#) that China released a goal in 2023 for 9 GW_{ac} to begin construction by the end of 2025 and achieved half the target.

Annual Installed CSP in China (MW_{ac})



Cumulative Chinese CSP Capacity, by Technology



Agenda

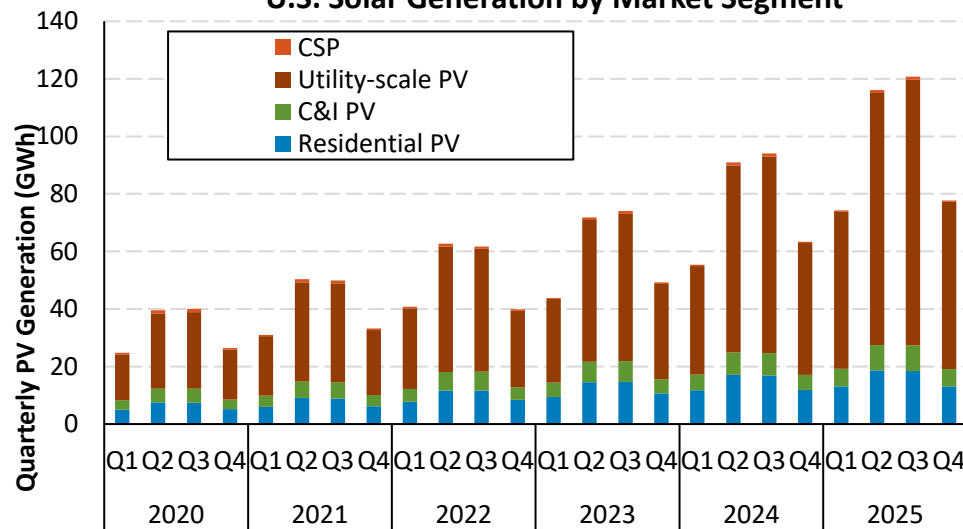
- 1 Global Solar Deployment
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- In 2025, the United States installed approximately 34.0 GW_{ac} of PV (SEIA reported 43.2 GW_{dc}), 15.2 GW_{ac} (41.9 GWh) of grid-scale storage, and 5.9 GW_{ac} of land-based wind.
 - PV installations were down 8% by capacity, y/y, with fewer large-scale projects.
 - EIA recorded another 2.1 GW_{ac} of distributed BESS.
- EIA projects that U.S. PV's annual growth will rise to 38.1 GW_{ac} in 2026 and 43.4 GW_{ac} in 2027. Other analyst estimates project 37 GW_{dc} to 44 GW_{dc} of annual PV deployment through 2029.

U.S. Solar Generation Breakdown (Quarterly)

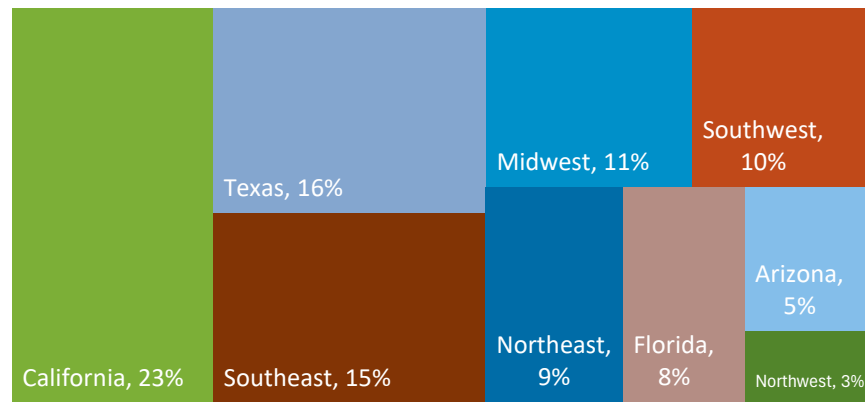
- The United States generated 368 TWh of solar generation in 2025, up 28% y/y.
 - Utility-scale PV generation grew 35%, followed by a C&I (13%), and residential PV (10%). CSP generation shrank 8%.

U.S. Solar Generation by Market Segment



- In 2025, more than half of PV generation occurred in California, Texas, Florida, and Arizona.
 - However, 31 states generated more than 2 TWh of electricity each from solar in 2025.

2025 U.S. PV Generation by Region (368 TWh)



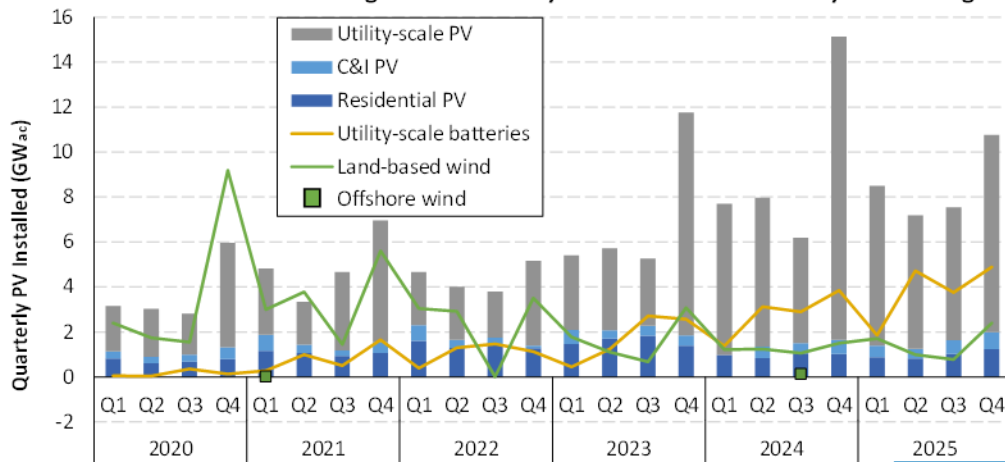
Note: "Northwest" includes Hawaii and Alaska.

Sources: EIA, [Electric Power Monthly](#), forms EIA-023, EIA-826, and EIA-861, Feb. 2026.

U.S. Installation Breakdown Quarterly: EIA (GW_{ac})

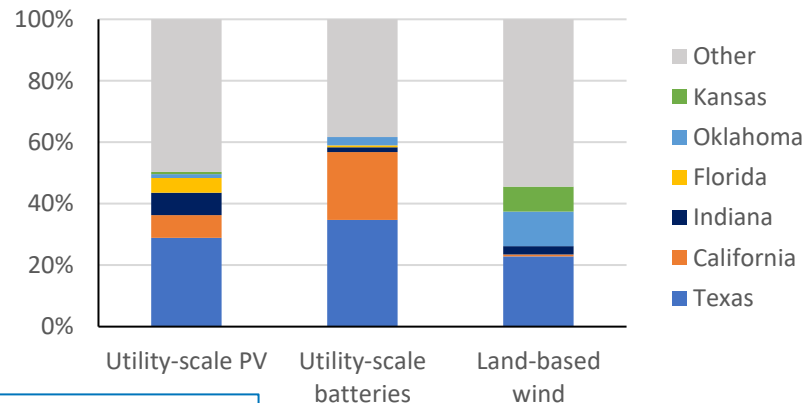
- It is estimated that the United States installed 34.0 GW_{ac} of solar capacity in 2025—down 8%, y/y.
 - The lower levels of deployment come from the downturn in large-scale PV deployment.
 - Approximately ½ of the utility-scale PV capacity installed in 2025 came from four states.

U.S. PV & grid-scale battery and wind installations by Market Segment



- The United States installed 15.2 GW_{ac} (41.9 GWh) of grid-scale battery storage in 2025—up 35% (37% by GWh).
 - Approximately 60% of the utility-scale battery capacity installed in 2025 came from two states.
- The United States installed 5.9 GW_{ac} of land-based wind in 2025—up 17%.
 - The installations occurred in 14 states, with Texas (28%), Oklahoma (14%), and Kansas (10%) leading the way.

Percentage of Capacity Installed in 2025, by State

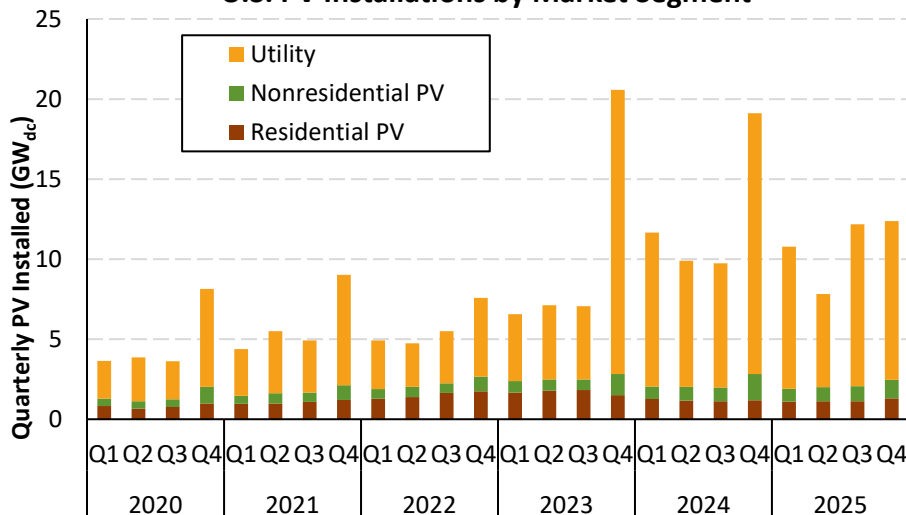


[Click here](#) to interactively view these data on Tableau Public.

U.S. Installation Breakdown Quarterly: SEIA (GW_{dc})

- Wood Mackenzie/SEIA reports 43.2 GW_{dc} of PV installations in 2025—a decrease of 14%, y/y.
 - Q4 2025 installations were down 35%, y/y.
 - Utility-scale PV and nonresidential PV installations, by capacity, were down 16% and 8% in 2025, respectively, from 2024. Residential PV installations in 2025 were down only 2% from 2024 but 32% from 2023.

U.S. PV Installations by Market Segment

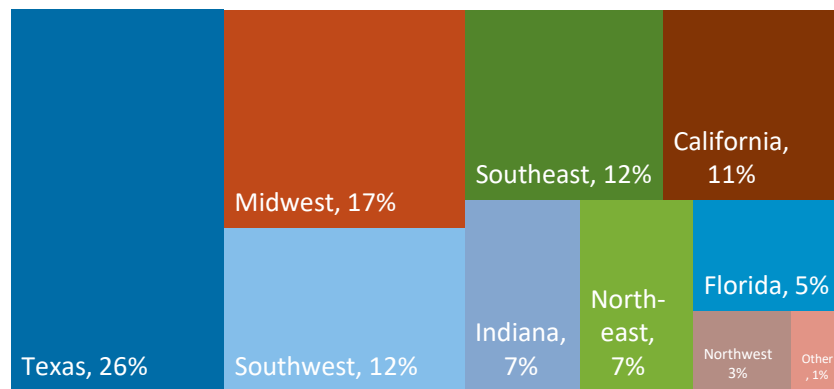


Note: Midwest excludes Indiana; Southeast excludes Florida; Southwest excludes Texas and California. **Source:** Wood Mackenzie/SEIA, [U.S. Solar Market Insight: 2025 Year-in-Review](#), Feb. 2026.

Unlike the previous slide, these values are in GW_{dc}—not GW_{ac}.

- SEIA reports that Texas installed approximately ¼ of PV capacity in 2025; however, 12 states installed more than 1 GW_{dc}.

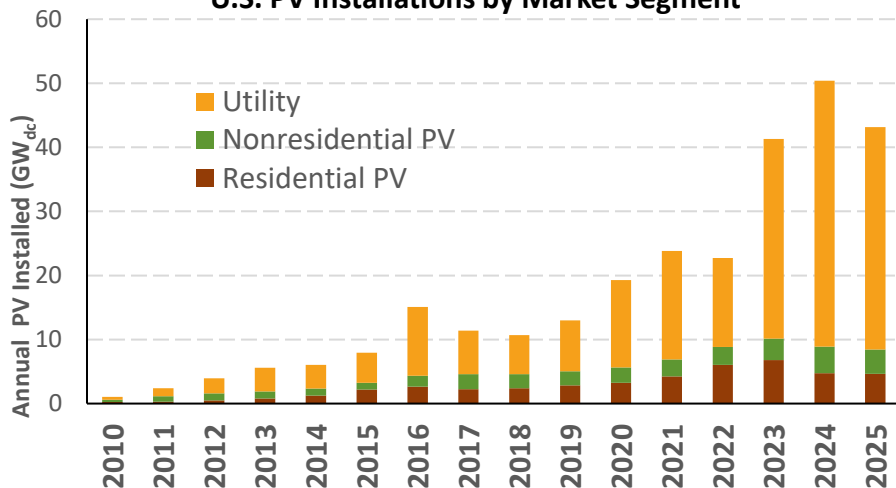
2025 U.S. PV Installations by Region
(43.2 GW_{dc})



U.S. Installation Breakdown Quarterly: SEIA (GW_{dc})

- According to Wood Mackenzie/SEIA, 2025 PV installations were the second-largest on record, bringing cumulative installations to 279.1 GW_{dc}.
 - Utility-scale PV represented 72% of cumulative U.S. PV installations at the end of 2025, followed by residential (16%) and nonresidential (12%).

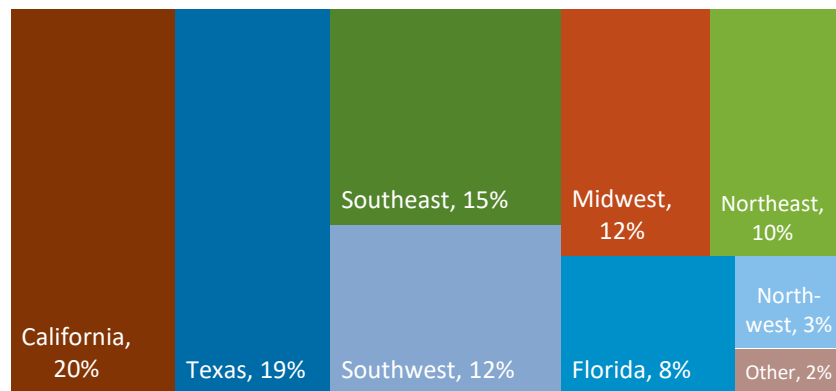
U.S. PV Installations by Market Segment



Unlike the previous slides, these values are in GW_{dc}—not GW_{ac}.

- SEIA reports that California (20%), Texas (19%), and Florida (8%) had the most PV capacity at the end of 2025.
 - At the end of 2025, 15 states had cumulative PV capacities above 5 GW_{dc}, and another 22 had more than 1 GW_{dc}.

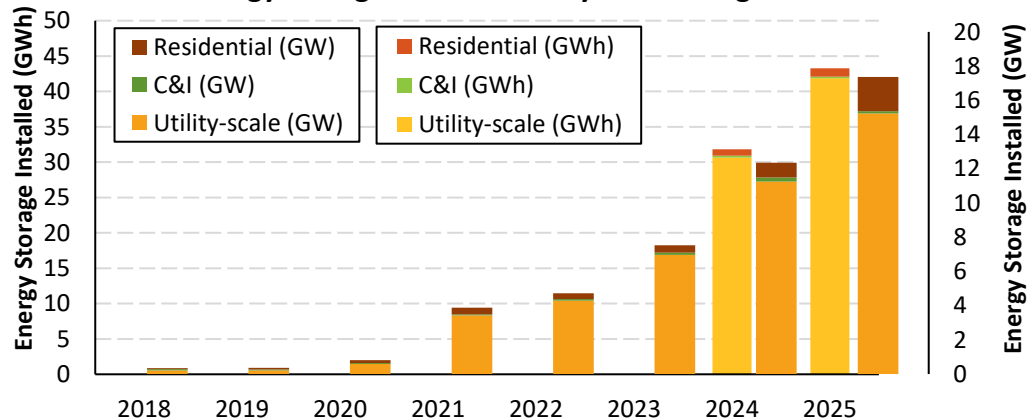
Cumulative U.S. PV Installations by Region (2025)
(279.1 GW_{dc})



U.S. Energy Storage Installations by Market Segment (EIA)

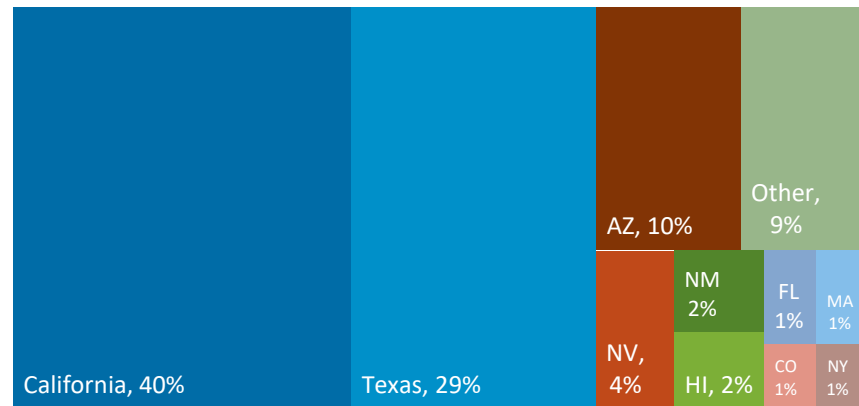
- EIA reports that the United States installed approximately 17.4 GW_{ac} (43.3 GWh) of energy storage onto the electric grid in 2025—up 36%–40%, y/y—bringing cumulative battery capacity to 48.0 GW_{ac} (121.2 GWh).
 - Approximately 89% of EIA-tracked battery capacity was utility-scale, followed by residential (9%).

U.S. Energy Storage Installations by Market Segment



- Like PV, BESS largest markets were California and Texas, followed by other southwestern states.
- Twenty-four states had installed more than 100 MW_{ac} of battery storage by the end of 2025.

2025 Cumulative U.S. Energy Storage Installations by Region (48.0 GW)

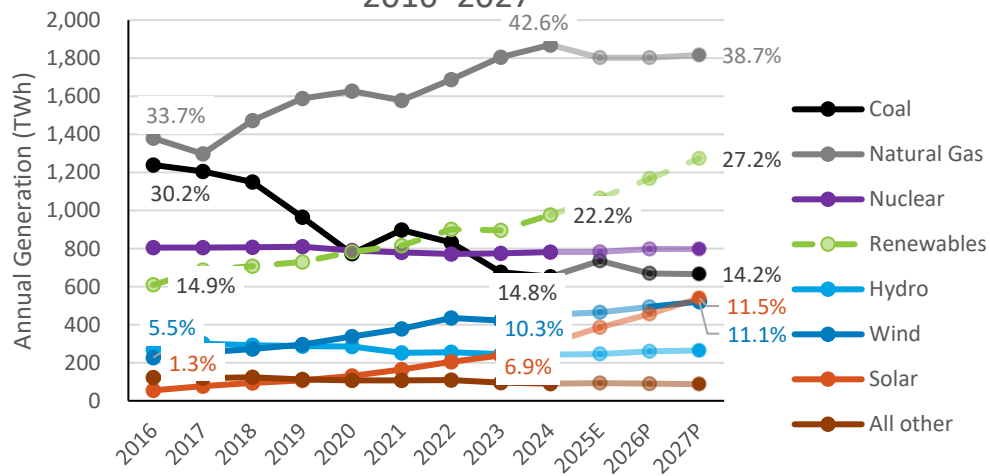


U.S. Generation by Source: 2016–2025 and Planned 2026–2027

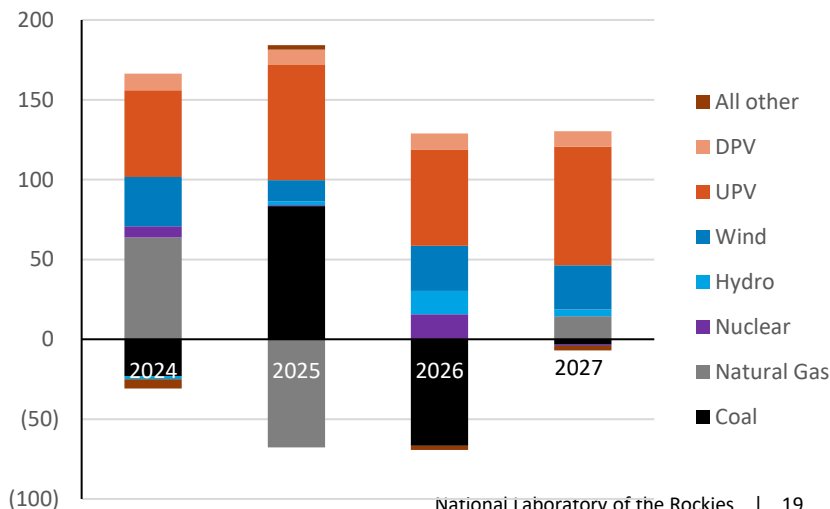
- From 2016 to 2024, solar, natural gas, and wind generation in the United States increased the most, by technology.

- EIA projects that majority of load growth in the United States in 2026 and 2027 will be met by solar electricity generation, growing from 6.9% of total U.S. electricity generation in 2024 to 11.5% in 2026.

Annual Electricity Generation by Technology, 2016–2027

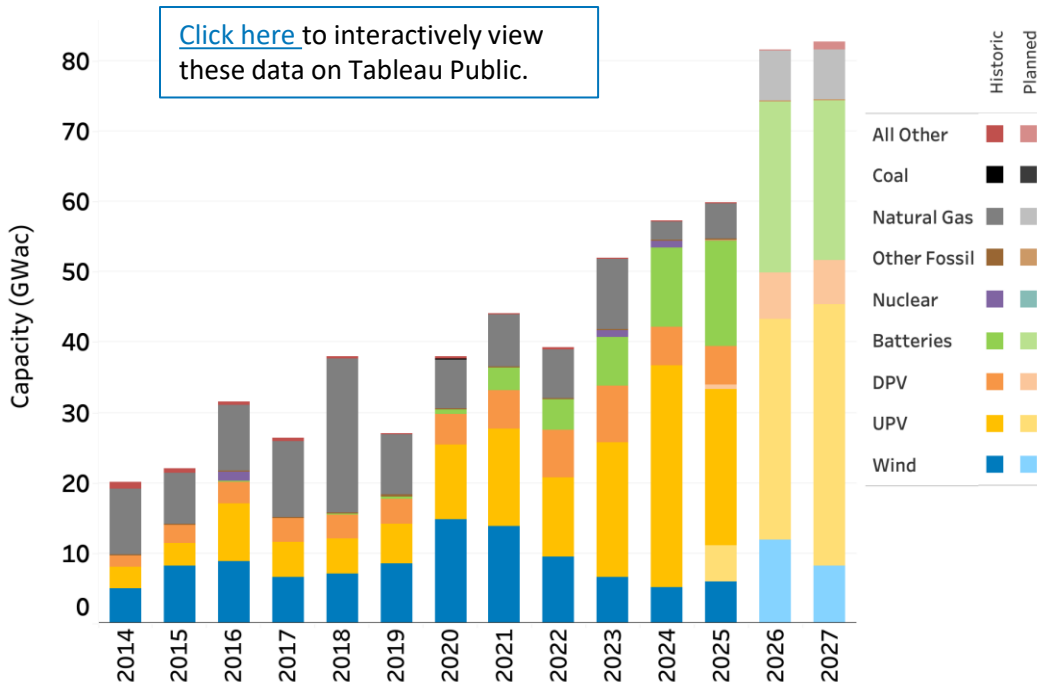


Change in Electricity Generation by Technology (TWh), 2024–2027



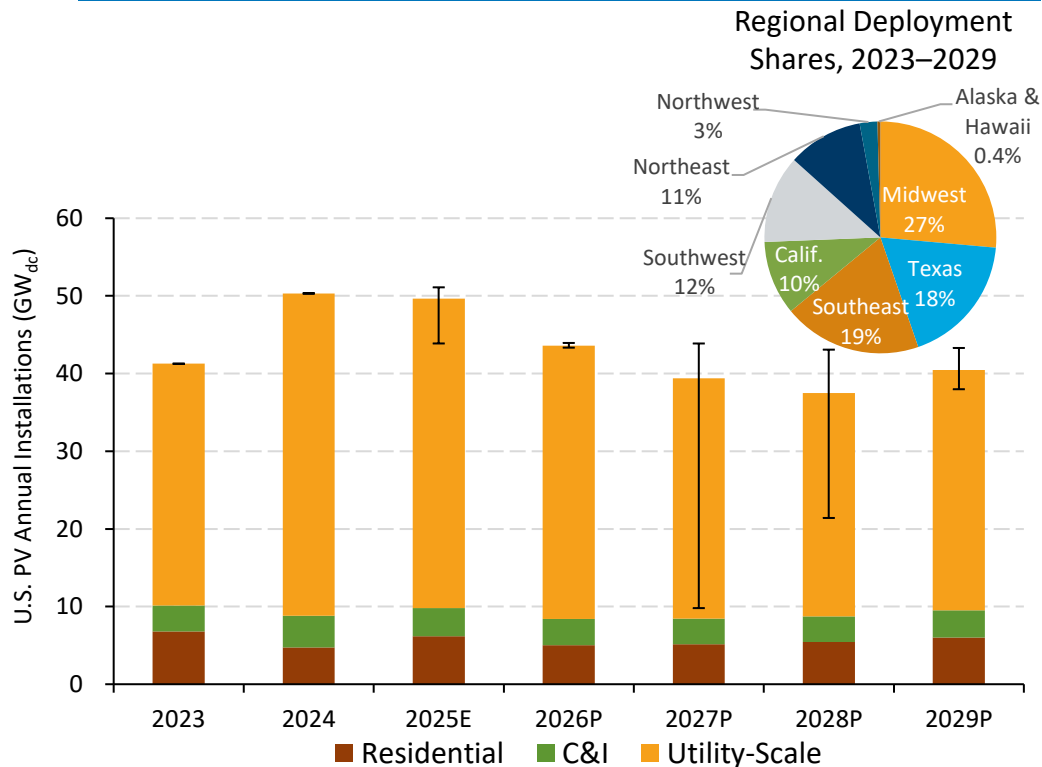
U.S. Generation Capacity Additions

Source: 2014–2025 and Planned 2026–2027



- EIA estimates that U.S. PV's annual growth will rise from 34.0 GW_{dc} in 2025 to 38.1 GW_{ac} in 2026 and 43.4 GW_{ac} in 2027.
 - EIA also estimates additions of 23 GW_{ac} of grid-scale BESS and 10 GW_{ac} of wind per year, in 2026 and 2027. This means over the next 2 years there would be 148.7 GW_{ac} of added PV, wind, and grid-scale storage.
 - EIA's estimate for 2025 PV, wind, and BESS additions is 13% lower than its projection from February 2025; however, its projection for additions from those technologies for 2026 is now 30% higher.
- EIA estimates that 19.4 GW_{ac} of natural gas capacity will be added between 2025 and 2027.
 - EIA began projecting into 2027 in February 2026, so there are no earlier 2027 projections for comparison; however, EIA projections from February 2026 increased the 2025 and 2026 projections for natural gas additions by 10% and 31%, respectively, when compared to the projections from February 2025.

Annual U.S. PV Deployment

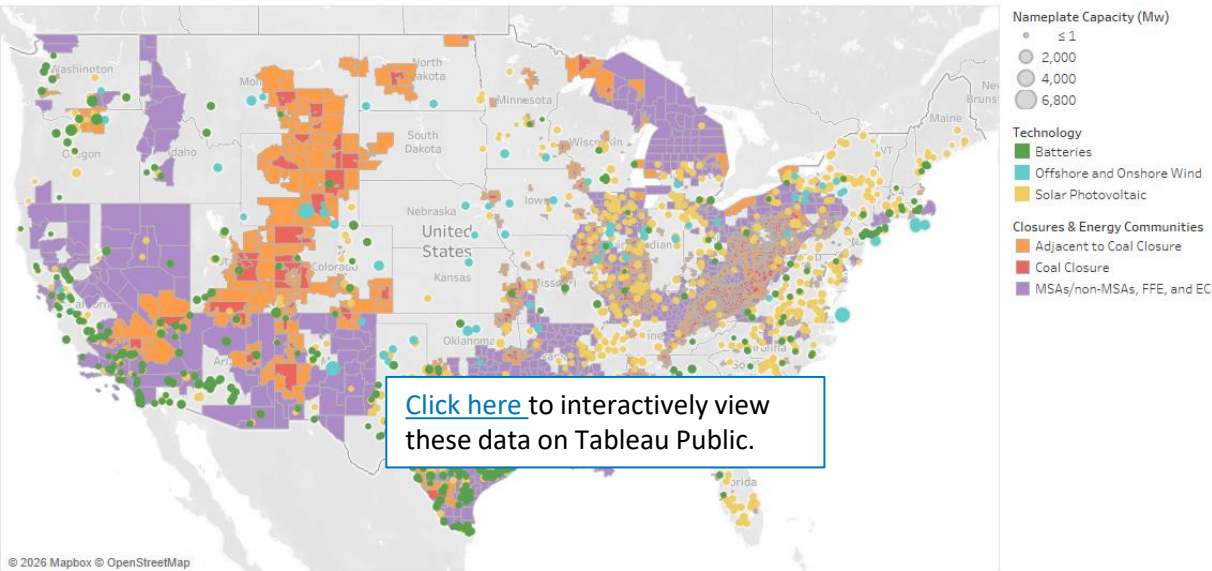


- Annual deployment of PV in the United States stayed about the same between 2024 and 2025, at 50 GW_{dc}.
- Analyst projections suggest a generally declining near-term trend in annual U.S. PV installations:
 - 44 GW_{dc} in 2026 (–12% y/y)
 - 39 GW_{dc} in 2027 (–10% y/y)
 - 37 GW_{dc} in 2028 (–5% y/y)
 - 40 GW_{dc} in 2029 (+8% y/y).
- Annual utility-scale installations shrink by –6%/y on average, while the distributed sectors average –1%/y.
- Analyst estimates of near-term installations fell over the past year largely because of U.S. policy changes.
 - For example, compared with projections made in late 2024, projections made in late 2025 show installations declining 13%–32% in 2028.
 - Passage of the “One Big Beautiful Bill Act” in July 2025 created uncertainty about project economics by changing timelines and eligibility for receiving solar tax credits.

Notes: E = estimate; P = projection. Bars represent median U.S. projections. Error bars represent high and low U.S. projections. Where sector medians do not sum to U.S. medians, the differences are reconciled by multiplying the median percentage contribution from each sector by the total U.S. median values, so the correct total U.S. median values are retained. **Sources:** BNEF, 4Q 2025 Global PV Market Outlook, Dec. 16, 2025; EIA, [Annual Energy Outlook 2025](#), April 2025; Wood Mackenzie and SEIA, US Solar Market Insight Q4 2025, Dec. 2025.

Planned U.S. Electric Generation Plants in Relation to Energy Communities

Planned U.S. Generation Capacity in Relation to Energy Communities, October 2025

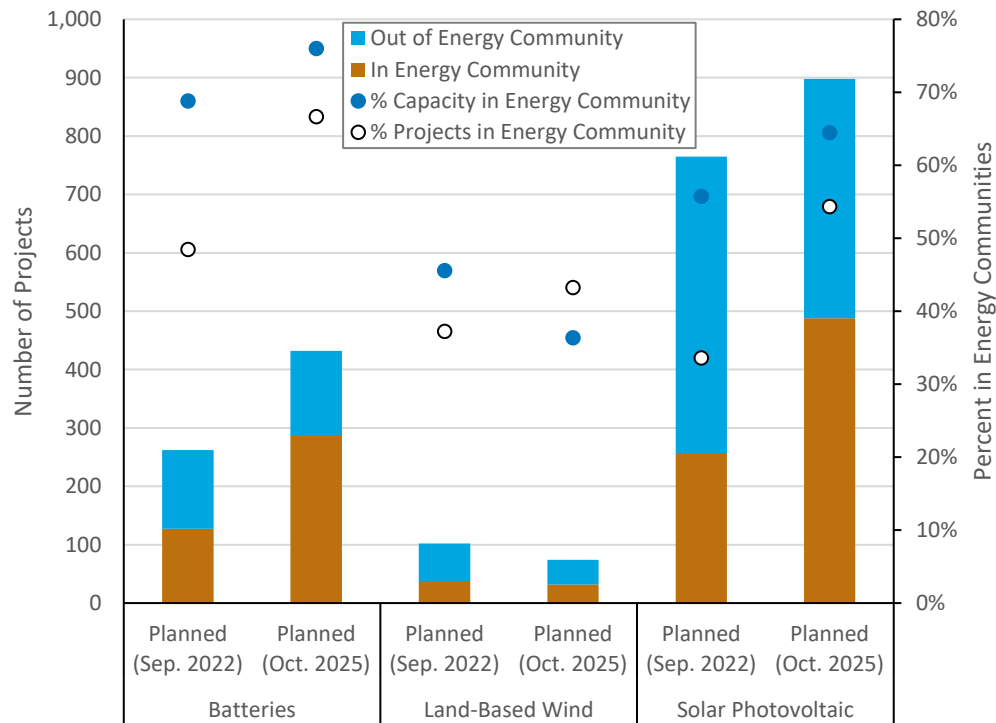


- Since “Energy Communities” were established in 2022, a significant number of new planned renewable energy projects have been announced.
- In September 2022, 1 month after the Energy Communities bonus was established, the EIA tracked over 1,100 planned utility-scale PV, land-based wind, and battery projects, of which 37% were located in energy communities (based on current guidelines).
 - In October 2025, there were 1,400 projects, of which 58% were in energy communities.

Note: Brownfields, which are also eligible for the energy community bonus credit, do not have the same census tract delineations and so are not included in this analysis. There is some uncertainty about exactly where a planned project will be built. This analysis relies on the latitude and longitude provided to EIA. **Source:** U.S. Energy Information Administration (EIA), EIA Form 860 (Nov. 2022, Nov. 2024).

Percentage of Planned PV, Wind, and Battery Projects Located in Energy Communities

[Click here](#) to interactively view these data on Tableau Public.



- NLR analysis found that more planned **projects**, across technologies, were located in energy communities in October 2025 than were in the pipeline in September 2022.

- In October 2025, approximately 54% of PV planned projects above 1 MW in size were located in an energy community, compared to 34% in 2022 (and 44% in September 2024).

- The percentage of **capacity** in energy communities was higher than the percentage of **projects** in energy communities for PV and batteries, but the growth from September 2022 to October 2025, in terms of capacity, was not as great.

- In October 2025, approximately 64% of PV planned capacity (from systems above 1 MW in size) were located in an energy community, compared to 56% in 2022 (and 60% in September 2024).

- Energy community market share can be influenced more by large individual projects when measured in terms of capacity rather than in terms of projects.

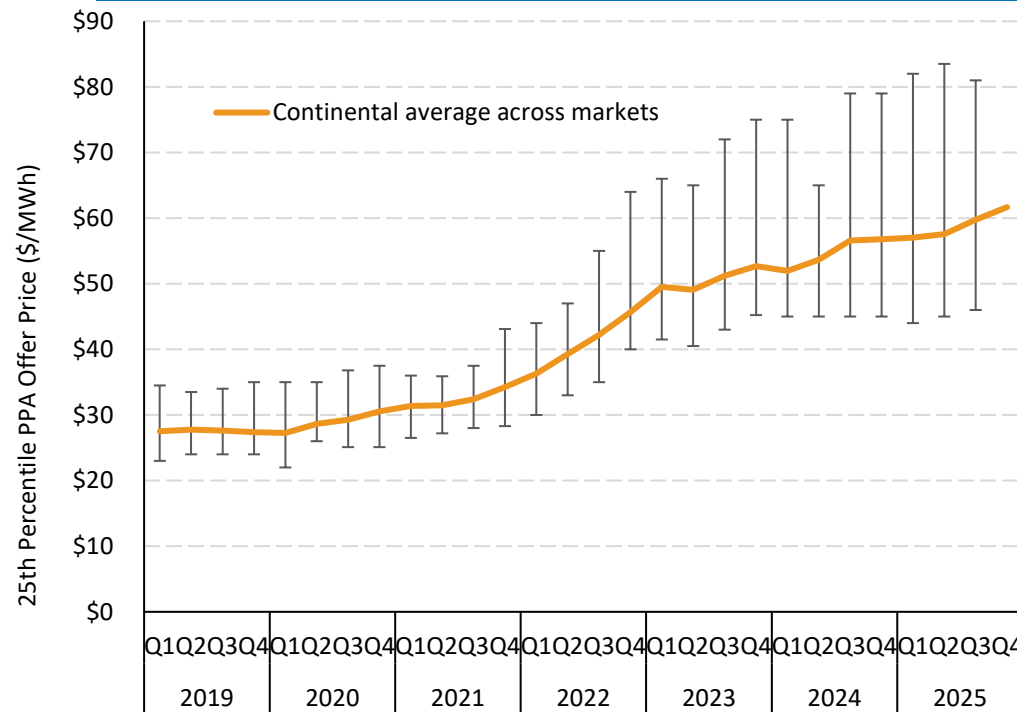
- Wind projects are more likely to claim a PTC, which provides less of an energy community bonus credit compared with the ITC; this may explain the lack of change in wind market size for energy communities.

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- 7 **U.S. PV Imports**

- LevelTen reports that, in Q4 2025, U.S. utility-scale photovoltaic PPA prices increased for the fifth quarter in a row, up 9%, y/y.
- From H2 2024 to H2 2025, the median real reported distributed PV system price (no storage) across AZ, CA, MA, and NY fell for smaller systems and generally rose for larger systems.
- Recent increases in Sunrun installation costs were due in part to increasing battery attachment rates, which grew from 27% in 2023 to 70% in 2025.

U.S. Solar PPA Pricing (LevelTen)



- LevelTen reports that, in Q4 2025, U.S. utility-scale photovoltaic PPA prices increased for the fifth quarter in a row, up 9%, y/y.
- LevelTen reported uncertainties for developers related to the Section 232 tariffs on aluminum, steel, copper, and polysilicon, new federal permitting procedures that have stalled some projects, pending regulations on FEOC compliance for tax credit qualifications, and not-yet-released revisions to Greenhouse Gas Protocol Scope 2 standards, which 97% of companies that track and disclose emissions use.

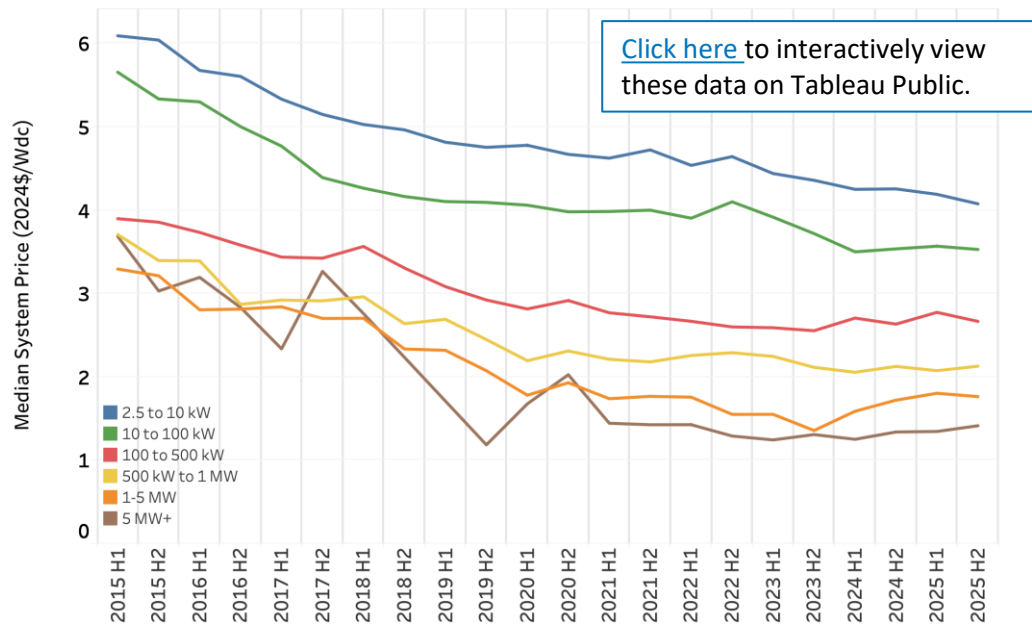
Note: In addition to U.S. wholesale markets (California Independent System Operator [CAISO], Electric Reliability Council of Texas [ERCOT], Southwest Power Pool [SPP], Midcontinent Independent System Operator [MISO], PJM, New York Independent System Operator [NYISO]; ISO New England), LevelTen also tracks pricing in Alberta, Canada (Alberta Electric System Operator [AESO]). **Source:** LevelTen, [Q3 2025 PPA Price Index Executive Summary North America](#), Aug. 2025.

Proposed Revision and Possible Impacts of Greenhouse Gas Protocol Scope 2

- [Greenhouse Gas Protocol](#) (GHGP) is a set of standards/guidelines for measuring and reporting GHG emissions. In general, it is voluntary and adopted by entities engaged in emissions reduction efforts. It is legally [required](#) under some programs—for example, in the EU.
- GHGP methods for reporting Scope 2 emissions (from energy use) include:
 - **Locational** uses average emissions intensity of energy delivered.
 - **Market-based** uses the emissions intensities of contracted supply (virtual PPAs, RECs).
- Proposed updates (phased in, exemptions based on data availability and entity size):
 - **Locational** would require using the most precise, publicly available geographic emission intensities at the most granular temporal level available.
 - **Market-based** would require hourly matching to load and contracts for resources in the same grid as the load, with additional protections against double-counting.
- GHGP is also considering a consequential accounting framework to model the emissions impact of investments separate from an entity's own energy use (e.g., for a project on a different grid).
- Potential impacts include:
 - More precise/accurate reporting (including revised progress); more additionality from investments
 - New markets for time and geography-specific energy attribute certificates
 - Increased complexity for adherence, less adoption (or increased abandonment) of GHGP.

Distributed PV System Pricing From Select States

State Distributed PV System Pricing (Aggregated biannually)



- From H2 2024 to H2 2025, the median real reported distributed PV system price (no storage) across AZ, CA, MA, and NY fell for smaller systems and generally rose for larger systems—see table.
- Across size ranges, real prices declined 31%–53% over the past 10 years and 8%–30% over the past 5 years.
- New York drove the recent price increases for systems of 1+ MW_{dc}.

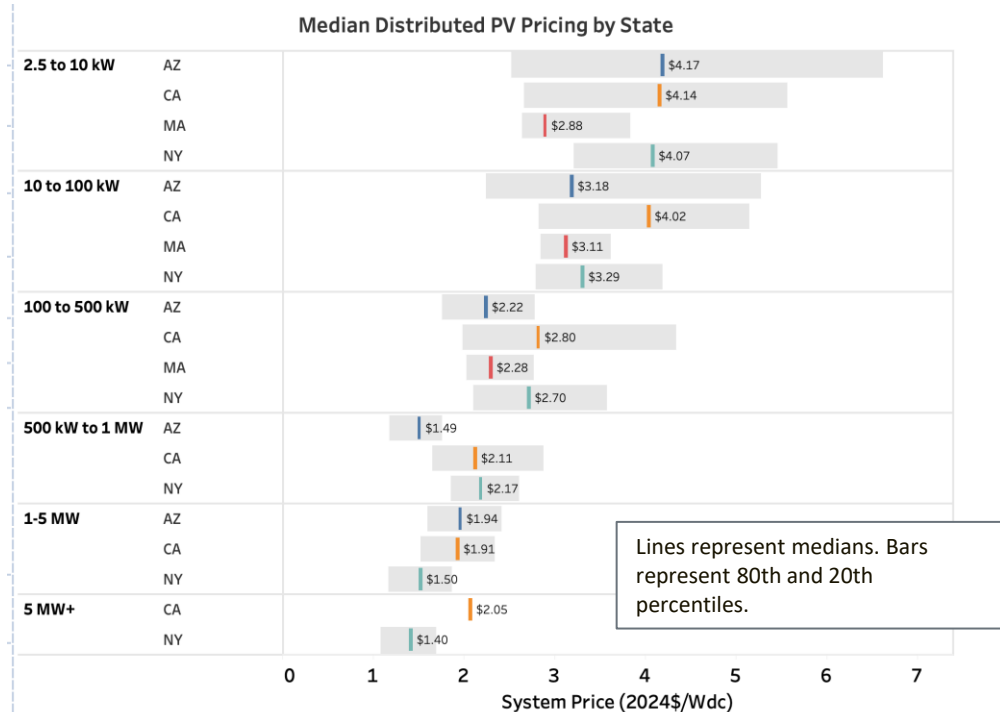
System Size (dc)	Median Price H2 2025 (\$/Wdc)	H2 2024 – H2 2025
2.5 to 10 kW	\$4.08	-4.2%
10 to 100 kW	\$3.53	-0.2%
100 to 500 kW	\$2.67	1.2%
500 kW to 1 MW	\$2.13	0.1%
1-5 MW	\$1.76	2.5%
5 MW+	\$1.41	5.6%

2025 MW_{dc}, year to date: Arizona (113), California (727), Massachusetts (8), New York (945).

Note: System prices above \$10/W and below \$0.75/W were removed from the dataset. The volatility in median system price among the largest systems is because of the relatively small number of systems deployed each year. **Sources:** Arizona Goes Solar, [Utility Programs](#), Feb. 2026; California Distributed Generation Statistics, [Distributed Generation Interconnection Program Data](#), [Interconnected Project Sites Data Set](#), Dec. 2025; Mass.gov, [Massachusetts Lists of Qualified Generation Units](#), Sept. 2025; Data.NY.gov, [Solar Electric Programs Reported by NYSERDA](#), Jan. 2026.

Distributed PV System Pricing, Select States, 2025 Year to Date

[Click here](#) to interactively view these data on Tableau Public.



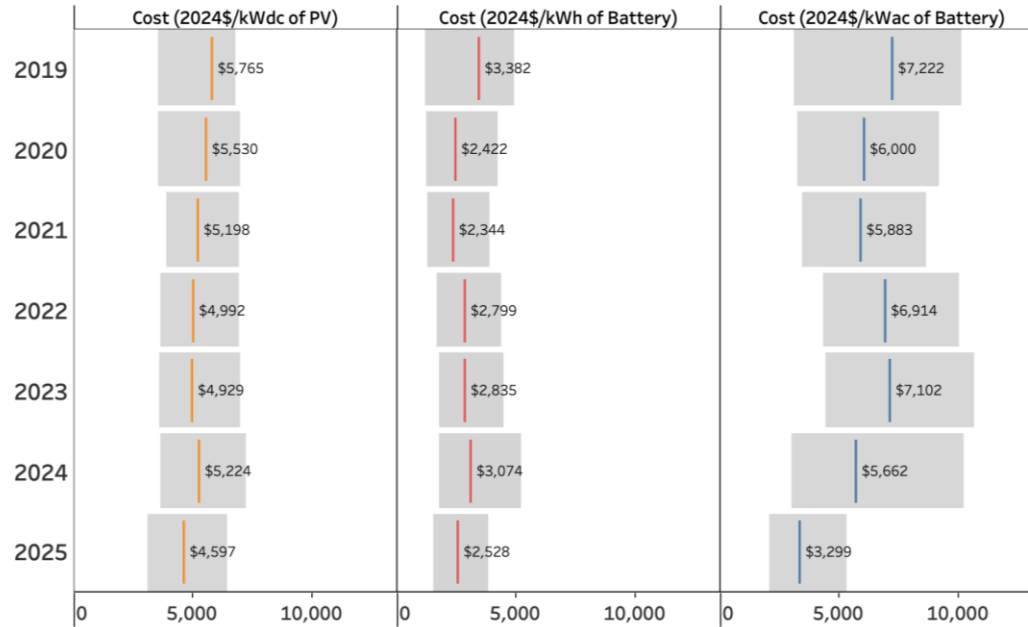
- In addition to price differences based on system size, there is variation in the price of stand-alone (no energy storage) distributed PV systems between states and within individual markets.
- Dollar-per-watt prices generally decrease as system size increases.
- For systems 2.5–10 kW_{dc}, median price changes varied between H2 2024 and H2 2025:
 - –9% in Arizona, –5% in California, and no change in New York.
 - Massachusetts data were unavailable in H2 2025.

2025 MW_{dc}, year to date: Arizona (113), California (727), Massachusetts (8), New York (945).

Note: System prices above \$10/W and below \$0.75/W were removed from the dataset. **Sources:** Arizona Goes Solar, [Utility Programs](#), Feb. 2026; California Distributed Generation Statistics, [Distributed Generation Interconnection Program Data](#), [Interconnected Project Sites Data Set](#), Dec. 2025; Mass.gov, [Massachusetts Lists of Qualified Generation Units](#), Sept. 2025; Data.NY.gov, [Solar Electric Programs Reported by NYSERDA](#), Jan. 2026.

Residential PV-Plus-Storage System Pricing in California

California Residential PV-Plus-Storage Pricing

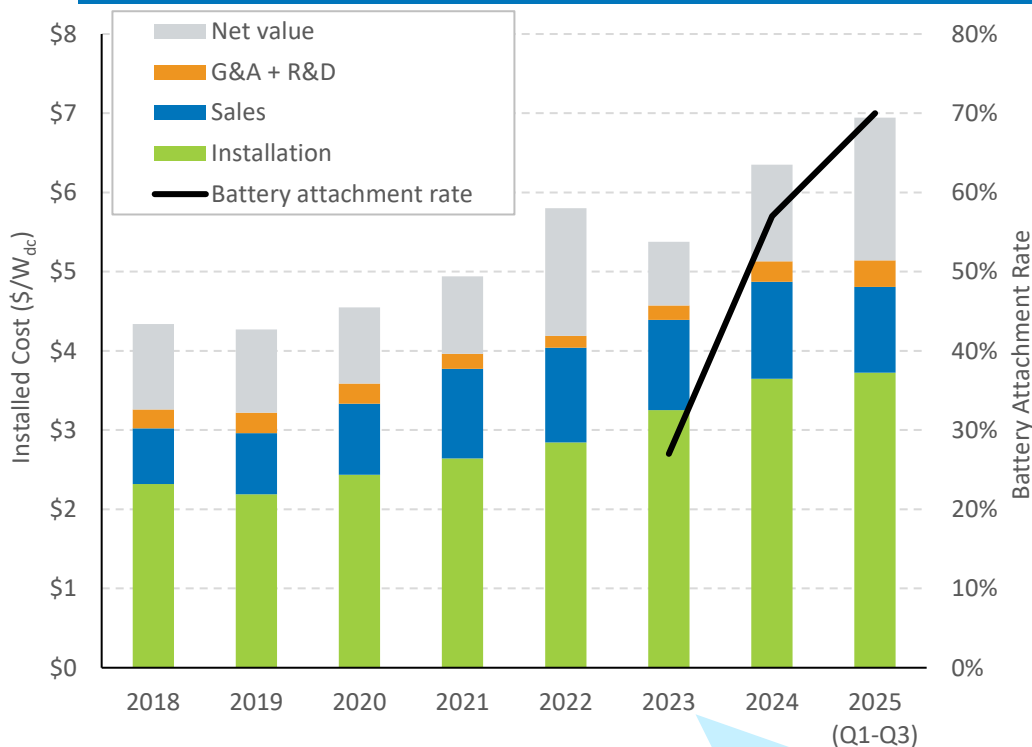


- In 2025, residential PV-plus-storage systems in California had a median system price of \$2,528/kW_{ac} of battery, \$3,299/kW_{ac} of battery, and \$4,597/kW_{dc} of PV.
 - The cost per kW_{ac} of battery fell 42% between 2024 and 2025, while the cost per kWh of battery fell 18% and per kW_{dc} of PV fell 12%.
- The increasing market share of Tesla’s Powerwall 3 likely contributed to the significant drop in PV-plus-storage price, especially in terms of kW_{ac} of battery.
 - Powerwall 3—which became available in 2024—provides the same **energy** as previous models (13.5 kWh) but provides 130% more continuous **power** (11.5 kW_{ac} vs. 5 kW_{ac}).
 - Tesla supplied 44% of California’s PV-plus-storage systems in 2024, rising to 55% in 2025.
 - The integrated inverter in Powerwall 3 serves the PV and battery components of a PV-plus-storage system, which eliminates the need for a separate PV inverter.

The data are filtered to residential rooftop PV systems between 3 kW_{dc} and 30 kW_{dc} with a cost between \$0.75/kW_{dc} and \$20/kW_{dc} (2024 dollars). Lines represent medians. Bars represent 80th to 20th percentiles.

Sources: California Distributed Generation Statistics, [Distributed Generation Interconnection Program Data](#), [Interconnected Project Sites Data Set](#), Dec. 2025; EnergySage, [Tesla Dominated the Home Battery Market—Will Its Reign Last?](#) May 2025; Tesla, [Powerwall](#), Sept. 2025; Wood Mackenzie, US Distributed Solar-Plus Storage 2025 Outlook, April 2025; Wood Mackenzie, US Distributed Solar-Plus-Storage Leaderboard, Dec. 2025.

Sunrun Cost and Value, 2018–2025



- National integrator Sunrun's cost of installing residential systems rose each year between 2019 and 2025, at an average annual rate of 9.3% (5.1% when adjusted for inflation).
- Recent increases in installation costs were due in part to increasing battery attachment rates, which grew from 27% in 2023 to 70% in 2025.
- Growth in subscriber value averaged 50% per year between 2023 and 2025, outpacing the increased installation costs.
 - Includes net present value of contracted cash flows, tax credits, and other benefits.
 - Value drivers include increasing storage attachment rates, increasing receipt of ITC bonus credits, ongoing cost efficiency efforts, and increasing sales of Sunrun's Flex product (introduced in mid-2025), which includes extra PV and battery capacity to meet future energy demand growth.

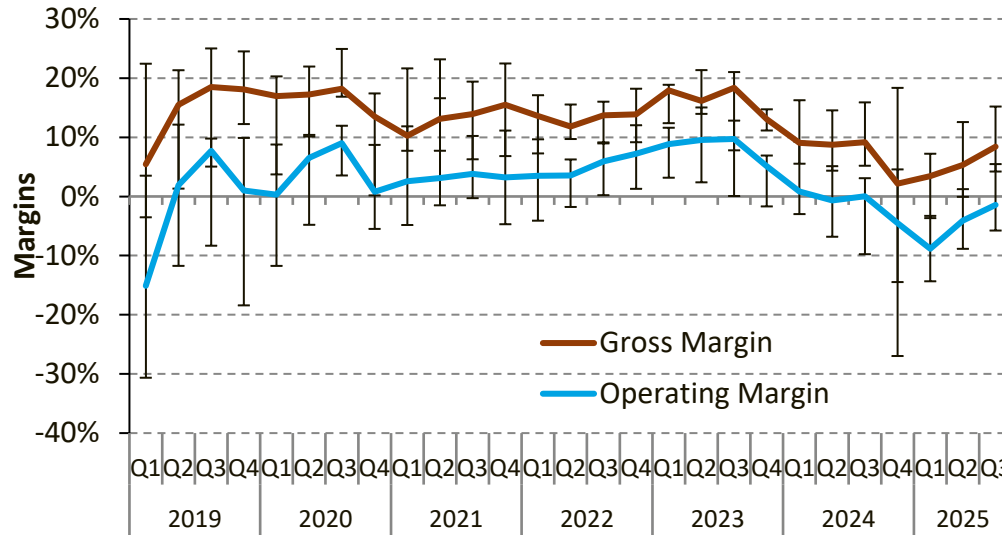
California's net metering policy incentivizes battery storage, ITC bonus credits become available

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- Margins for the leading PV wafer, cell, and module manufacturers continued to recover in Q3 2025, though most companies were still losing money.
- BloombergNEF reported that, at the end of 2025, the approximate global c-Si PV manufacturing capacity was between 1.35 and 1.67 TW—a growth of 2% (c-Si wafers) to 10% (c-Si modules)—with most still in China.
 - Due to manufacturing capacity built in 2025, at the end of the year, 40% of c-Si manufacturing capacity outside of China was in India.
- The United States manufactured approximately 24.9 GW_{dc} of PV panels in the first 9 months of 2025, up 103%, y/y, though flat, q/q.
 - At the end of Q3 2025, the United States had a c-Si cell manufacturing capacity of 3 GW_{dc}.

PV Wafer/Cell/Module Manufacturers' Margins



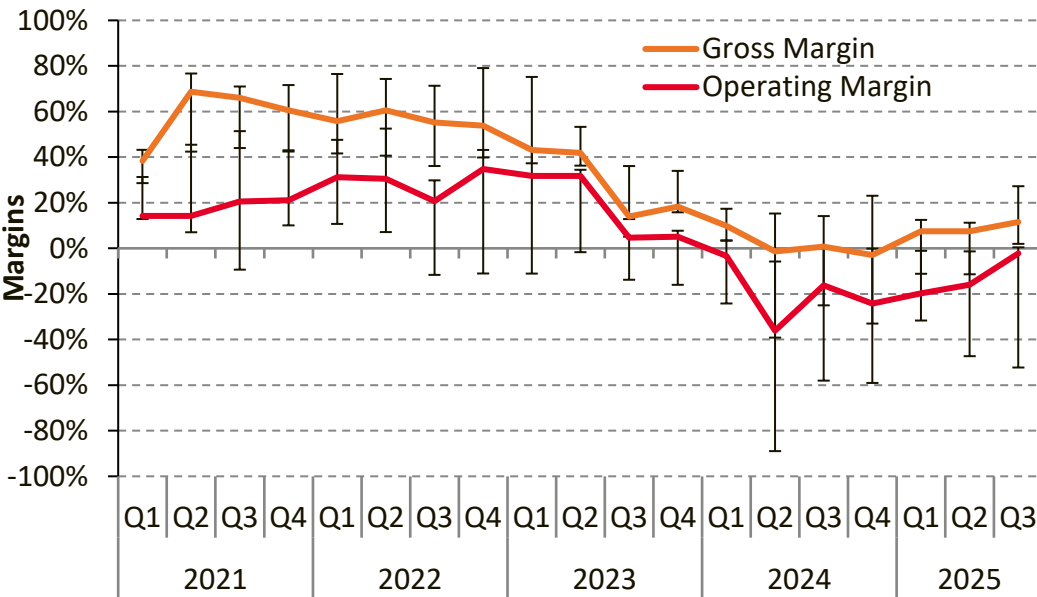
- Margins for the leading PV wafer, cell, and module manufacturers continued to recover in Q3 2025, though most companies were still losing money.
 - PVTech reported that LONGi, JinkSolar, and JA Solar had “significant” losses in Q3 2025 due to continued price declines and high levels of inventory.
 - First Solar lowered its guidance on 2025 profit because of (1) glass supply chain issues in Q3 (which are now resolved) and (2) a contract termination due to customer default.

Lines represent the median, with error bars representing the 80th and 20th percentiles for the following companies through Q3 2025: Canadian Solar, First Solar, GCL-Poly, Hainan Drinda New Energy Technologies, JA Solar, Jinko Solar, LONGi, Motech Industries, Risen, Shanghai Aiko Solar, Shanghai Aerospace, Tongwei, Trina Solar, and United Renewable Energy. Q3 2025 data are not available for Moxone and Renesola; however, the chart incorporates previous financial performance.

Note: Gross margin = revenue minus cost of goods sold (i.e., the money a company retains after incurring the direct costs associated with producing the goods or services it sells); operating margin = gross margin minus overhead and operating expenses (i.e., the money a company retains before taxes and financing expenses).

Sources: Company figures based on public filings and finance.yahoo.com and google.com/finance; PVTech, [Chinese PV Industry Brief: Major Solar Manufacturers Report Steep Q3 Losses](#), Nov. 2025.

PV Polysilicon Manufacturers' Margins



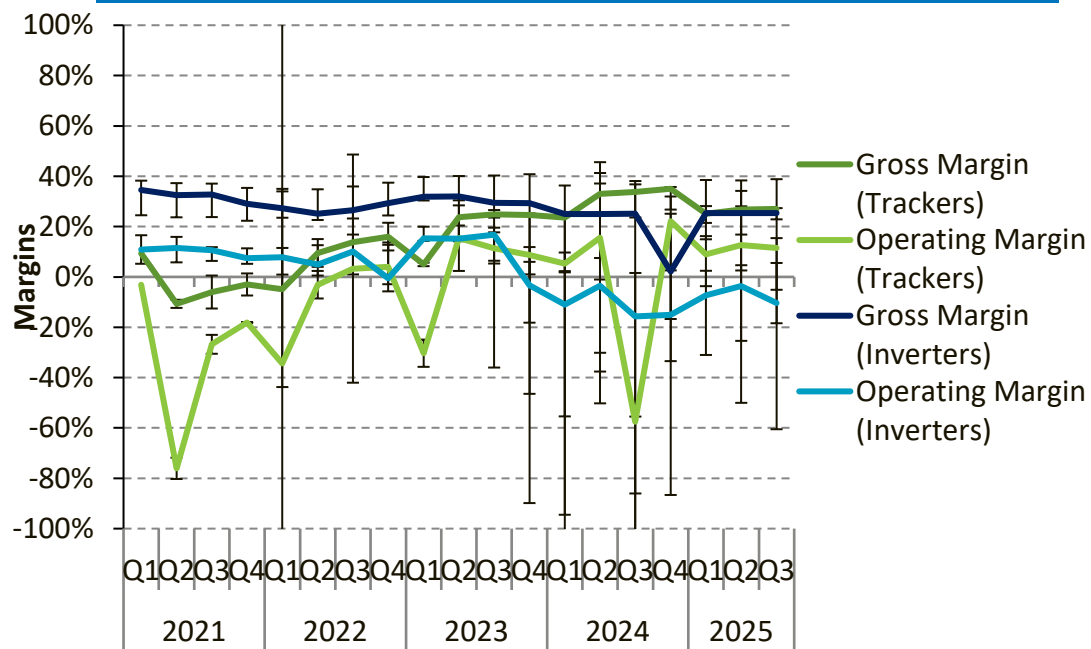
- Polysilicon manufacturers' margins continued to recover in Q3 2025 from recent lows in 2024; however, most manufacturers are still only breaking even.
 - Manufacturers attributed the turnaround to increased pricing and sales volumes.

Lines represent the median, with error bars representing the 80th and 20th percentiles for the following companies in Q3 2025: Daqo, OCI Holdings, REC Silicon, TCL Zhonghuan, and Wacker. GCL Poly and Xinte data only go through Q2 2025 because they report semi-annually.

Note: Gross margin = revenue minus cost of goods sold (i.e., the money a company retains after incurring the direct costs associated with producing the goods or services it sells); operating margin = gross margin minus overhead and operating expenses (i.e., the money a company retains before taxes and financing expenses).

Sources: Company figures based on public filings and finance.yahoo.com.

Inverter and Tracker Manufacturers' Margins



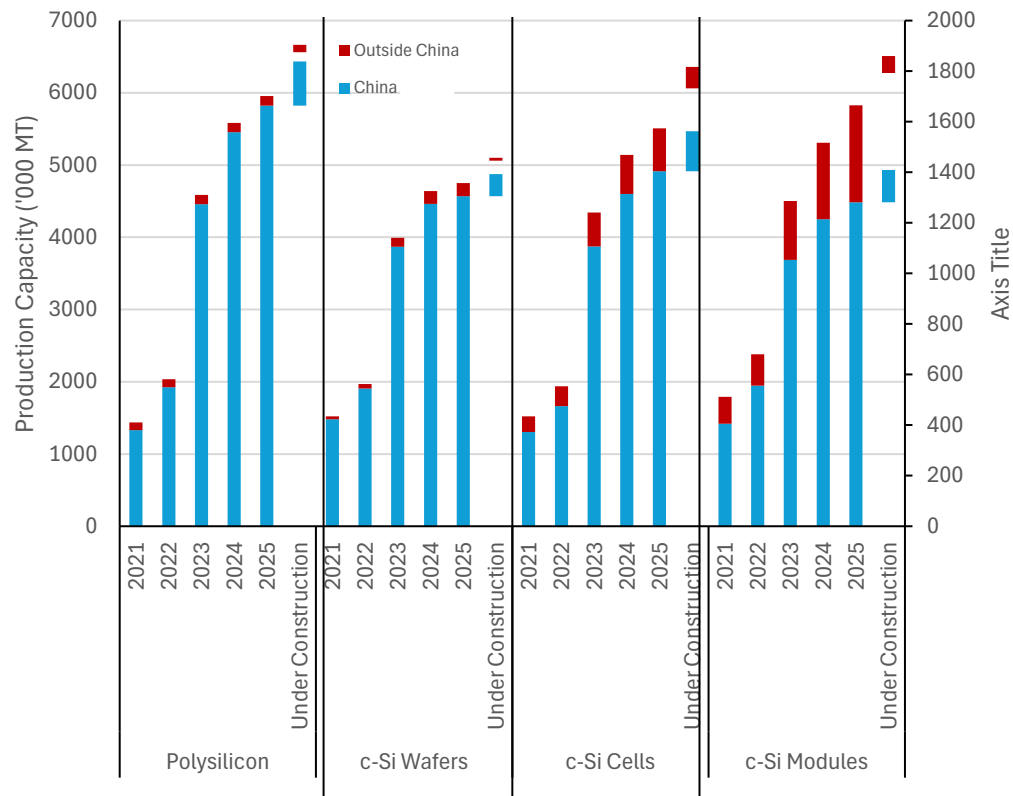
- Inverter and tracker manufacturers generally remained profitable through 2024 due to high levels of demand, though variation exists between companies and quarters.
- Some residential inverter companies began losing money in 2024 for a variety of reasons, including excess inventory, higher interest rates, and decreased demand in U.S. and Europe (i.e., NEM 3.0, reduced EU incentives post-Ukraine energy panic).
 - In Q3 2024, SolarEdge had a \$1 billion impairment write-down to correct the value of its assets.

Lines represent the median, with error bars representing the 80th and 20th percentiles for the following companies in Q3 2025: (trackers) Array Technologies, FTC Solar, NexPower; (inverters) Enphase, SMA, SolarEdge.

Note: Gross margin = revenue minus cost of goods sold (i.e., the money a company retains after incurring the direct costs associated with producing the goods or services it sells); operating margin = gross margin minus overhead and operating expenses (i.e., the money a company retains before taxes and financing expenses).

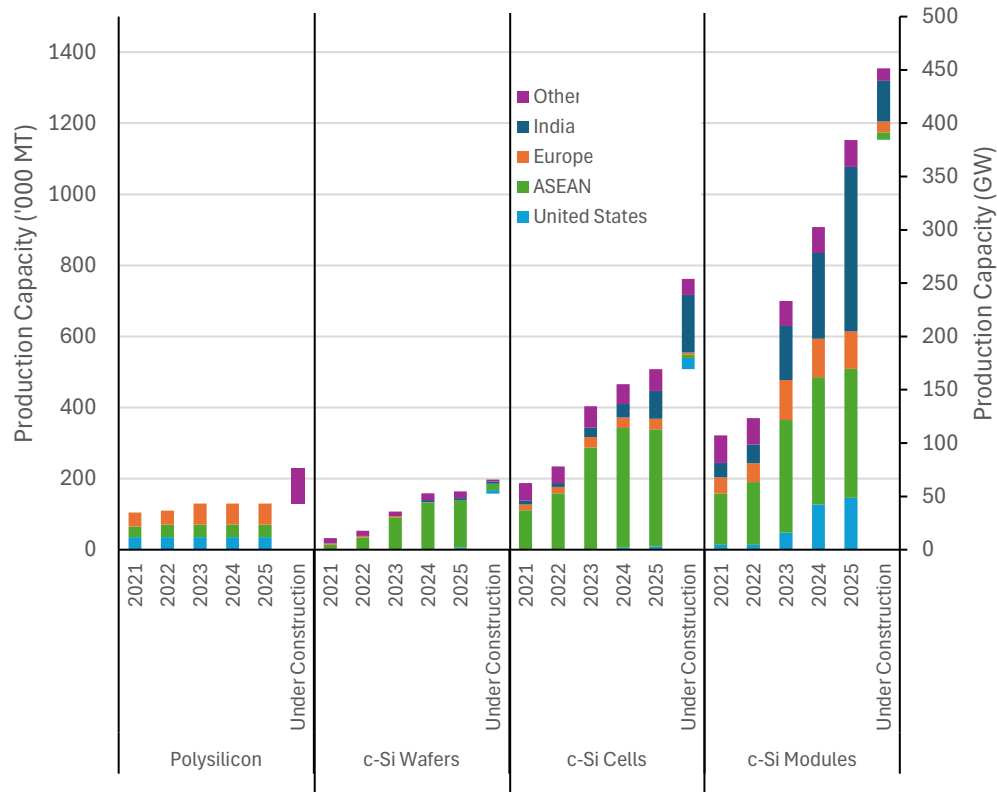
Sources: Company figures based on public filings and finance.yahoo.com.

Growth in Global PV Manufacturing Capacity



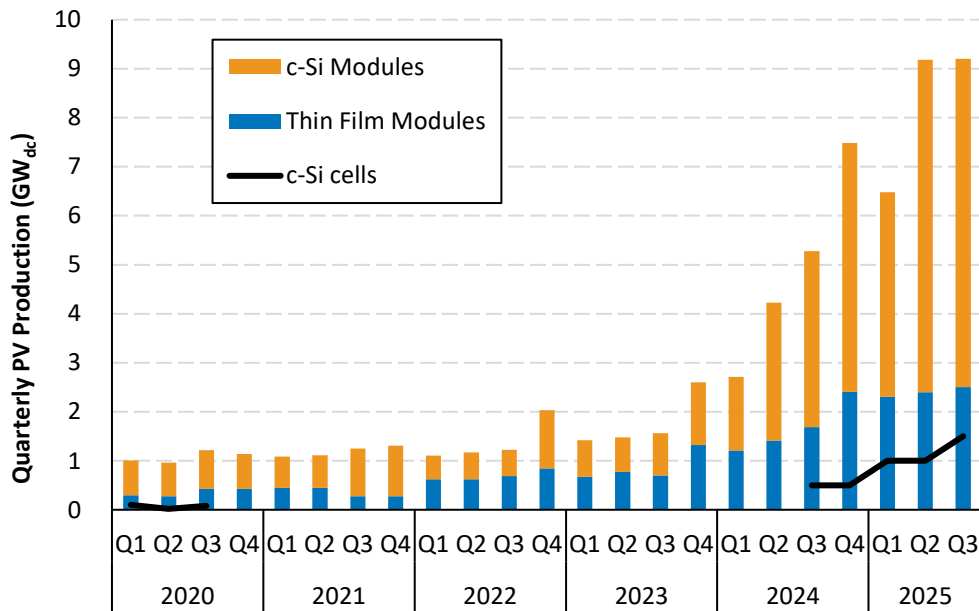
- BloombergNEF reported that at the end of 2025, the approximate global c-Si PV manufacturing capacity was between 1.35 and 1.67 TW—a growth of 2% (c-Si wafers) to 10% (c-Si modules).
 - The percentage of global manufacturing capacity in China increases up the supply chain (77% for c-Si modules compared to 98% for polysilicon).
 - Despite significant overcapacity, particularly since 2023, global PV manufacturing capacity continues to grow (though at a slower rate).
 - At the end of 2025, more than 100 GW of c-Si manufacturing capacity were under construction, mostly in China.

Growth in Global PV Manufacturing Capacity (outside China)



- BloombergNEF reported that, at the end of 2025, global manufacturing capacity outside of China increased further down the supply chain, with manufacturing capacity able to produce approximately 50 GW of polysilicon and almost 400 GW of c-Si modules.
 - At the end of 2025, 71% of the c-Si module assembly capacity outside of China was in India (40%) and ASEAN (31%), with the growth in 2025 coming from India.
 - Global c-Si cell and wafer capacity outside of China was predominantly in ASEAN (65% and 81%, respectively).
 - Polysilicon manufacturing capacity outside of China is located in the United States, Germany, and Malaysia, with a 100,000 MT facility under construction in Oman.

U.S. PV Manufacturing



- Wood Mackenzie reported the United States manufactured approximately 24.9 GW_{dc} of PV panels in 9 months of 2025, up 103%, y/y, though Q3 2025 was flat, q/q.
 - Domestic c-Si PV cell production has begun to ramp, with Q3 2025 manufacturing capacity reaching an estimated 3 GW_{dc} of annual production.

Note: c-Si assumes capacity utilization of 50%.

Source: Wood Mackenzie/SEIA, [U.S. Solar Market Insight: Q4 2025](#), Dec. 2025.

Domestic Manufacturing Updates

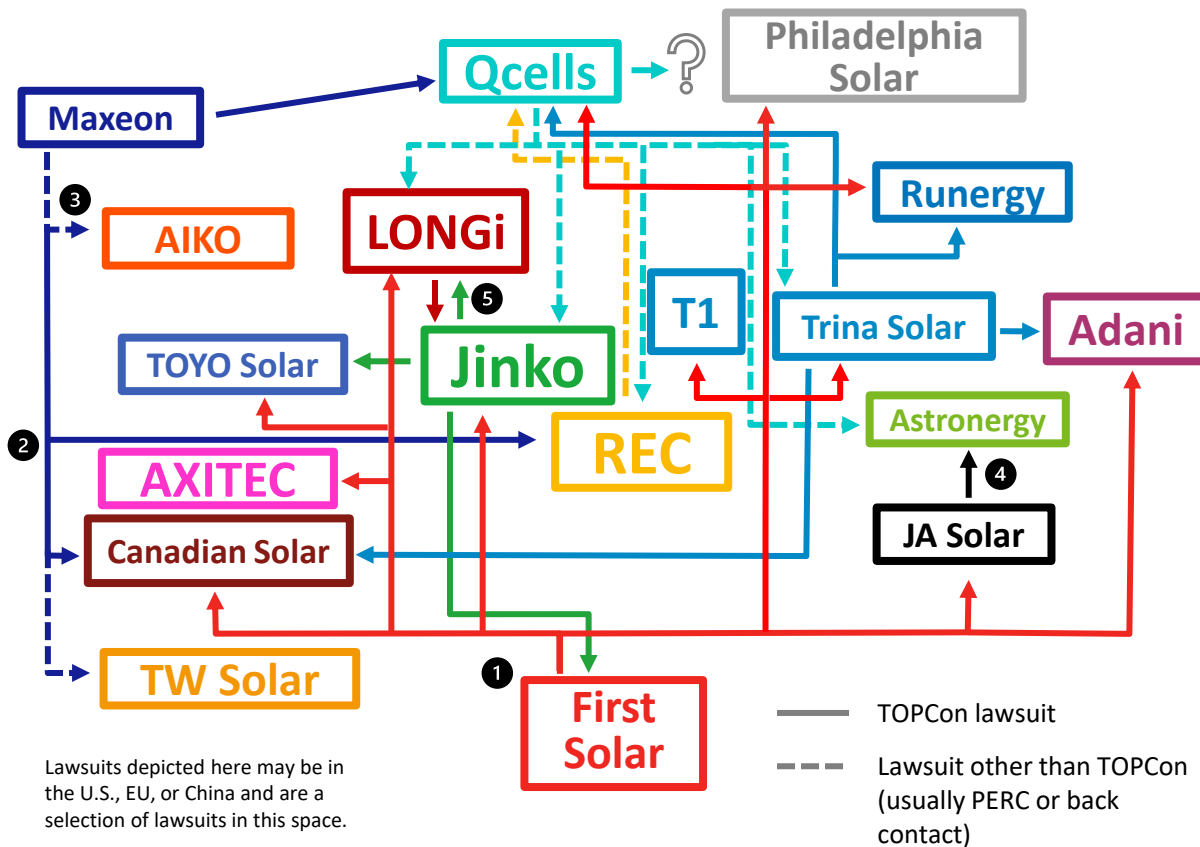
Some notable recent activities include:

- **T1 Energy** made several moves at the end of 2025/early 2026:
 - In December, T1 concluded a series of transactions to divest from Trina Solar to maintain its tax credit eligibility.
 - T1 also signed a 3-year supply deal with Treaty Oak Clean Energy for 900 MW of solar modules made from U.S. cells.
 - In January, T1 sold \$160 million worth of solar module manufacturing tax credits.
- **Tesla** made several moves in January 2026:
 - [Tesla announced](#) it was launching a U.S.-made solar panel from its Gigafactory New York.
 - Elon Musk announced that SpaceX (100 GW) and Tesla (100 GW) will construct 200 GW of solar capacity in 3 years.

Some notable recent activities include:

- Several companies announced supply deals:
 - **Toyo Solar** announced a 1-year polysilicon supply agreement with an undisclosed U.S. company.
 - **Jinko Solar** and **Nextpower** (formerly Nextracker) signed a deal for steel frames.
 - **NexWafe** and **Talon PV** announced a supply agreement for wafers.
- **Stewart Glass** received a \$16.4 million loan from the USDA for their solar glass facility in Ohio starting in March 2026. In contrast, **Canadian Premium Sand** has paused its plans to build a solar glass manufacturing facility in the southern United States.
- **SolarEdge** shipped U.S.-made inverters to Europe (it shipped to Australia in Q3 2025).
- Similar to T1, **Toyo** and **Canadian Solar** also both completed ownership acquisitions of their domestic manufacturing facilities, likely to comply with the foreign ownership rules of OBBBA.

PV Cell Architecture Lawsuits



Recent News

- 1 First Solar** – in January, the U.S. Patent and Trademark Office denied three separate attempts by JinkoSolar, Canadian Solar, and Mundra Solar (owned by Adani) to invalidate First Solar’s TOPCon patents, but the full case is yet to be heard. In March, First Solar filed a Sec. 337 request with the U.S. International Trade Commission against 10 competitors.
- 2 Maxeon** – in January, the U.S. Patent and Trademark Office ruled that 2 of Maxeon’s 3 patents were “unpatentable” and the third inconclusive. These patents were the basis of their lawsuits against Canadian Solar, REC, and Qcells.
- 3 Aiko** – in February, Aiko and Maxeon reached a licensing agreement on back-contact technology.
- 4 JA Solar/Astronergy** – in December, Astronergy and JA Solar agreed to end TOPCon litigation worldwide.
- 5 JinkoSolar/LONGi** – in September, LONGi and JinkoSolar agreed to a global settlement of all their ongoing patent disputes.

Sources: [US Patent Office Validates First Solar's TOPCon Intellectual Property, First Solar Asks US To Review Alleged TOPCon Patent Infringement by 10 Competitors](#), [US Patent Office Rules Two Maxeon TOPCon Patent Claims as “Unpatentable,” JinkoSolar and LONGi Settle Patent Disputes, But PV “Patent War” Continues](#), [Maxeon and Aiko Sign Global Patent Licensing Agreement for Back Contact Solar Technology](#), [Astronergy, JA Solar Terminate Patent Disputes, Agree on Licensing](#).

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- Global polysilicon, wafer, and cell pricing increased 30%–44% in 2025, attributed to Chinese government intervention and the rising price of silver. Global module prices were around $0.10/W_{dc}$ at the end of January 2026.
- The U.S. solar supply chain, however, is only indirectly impacted by Chinese market conditions due to tariffs and other trade restrictions; module prices in the United States remained around $\$0.27/W_{dc}$ at the end of January 2026.
 - Import value of PV modules decreased dramatically in H2 2025; however, tariffs are likely keeping the price-to-consumer at relatively steady levels.
- After a small increase in 2022, stationary storage battery pack prices continued their historical downward trend in 2025, falling 84% from 2016 to 2025 and 45% between 2024 and 2025, to $\$70/kWh$.

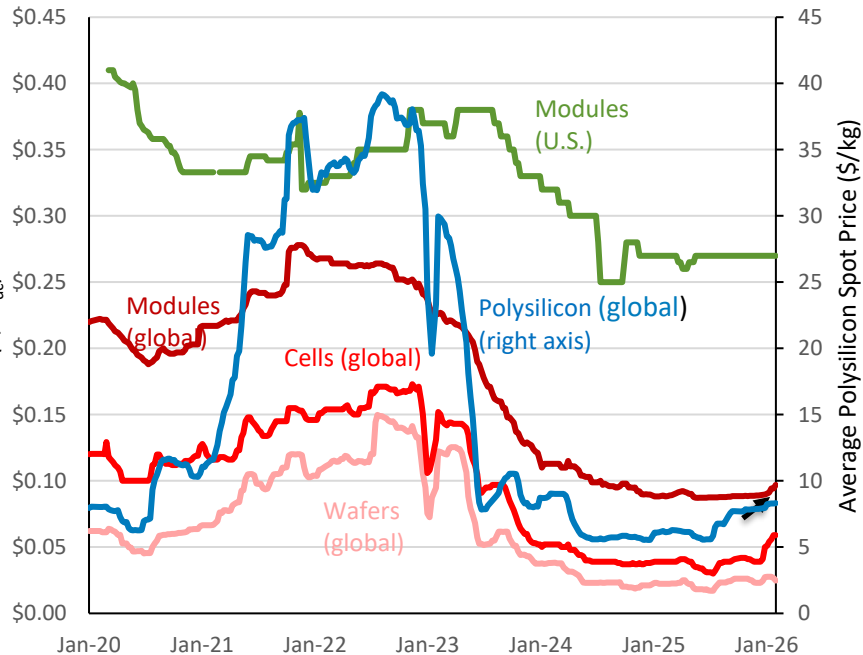
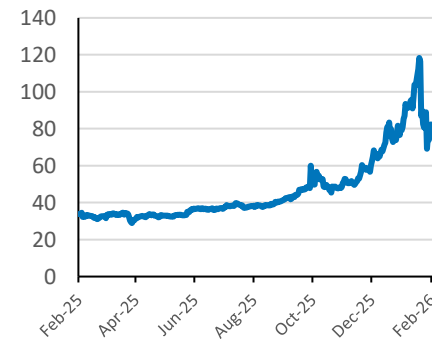
Recent Global PV Component Price Changes

- Global polysilicon, wafer, and cell pricing increased 30%–44% in 2025, while c-Si module pricing ended where it began.
- c-Si cell and module pricing continued to rise in January 2026, increasing 18% and 8%, respectively.
- U.S. module prices continue to trade at more than 2.5X global prices.

Several factors influenced the increase in pricing:

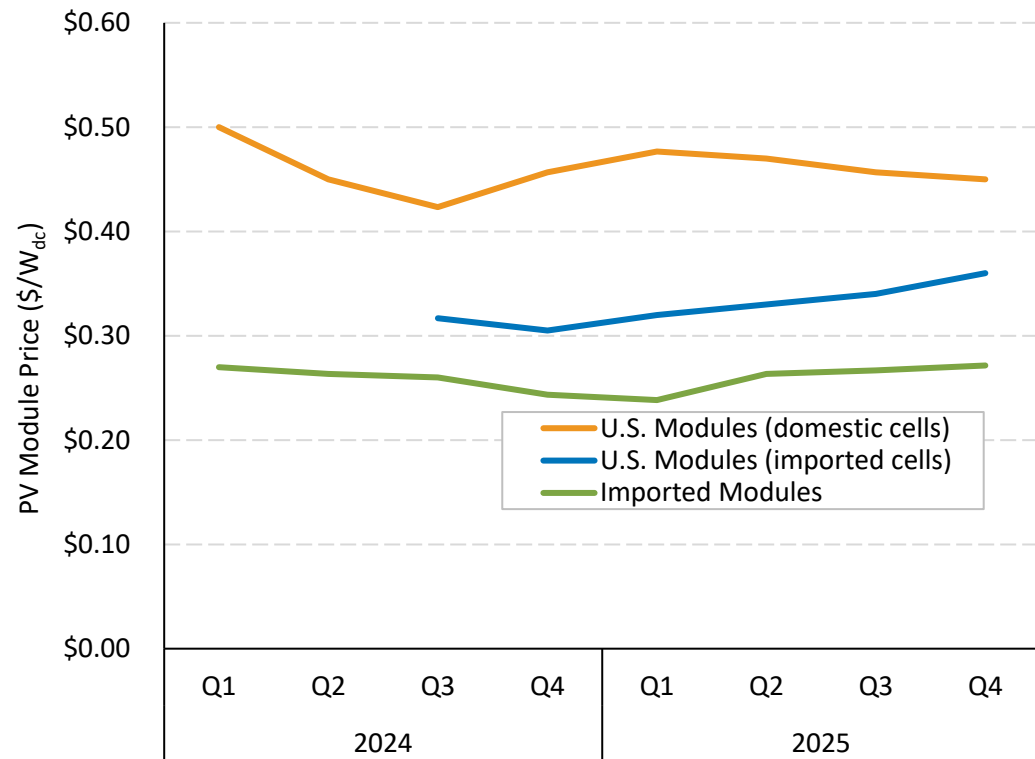
- The increase in pricing was attributed to [new Chinese government policies](#) intended for more sustainable pricing.
- [PV Magazine reported in January 2026](#) that the rising cost of silver over the past 6 months has put pressure on solar manufacturers, where silver paste accounted for 30% of total cell production costs—greater than polysilicon or any other component.
 - In late January 2026, silver prices dropped at a near-record rate, which was attributed to a strengthening dollar and investors seeking to cash in on some of the increased value.
 - Even after the drop in the price of silver, it is still more than double the price in mid-2025.
- In early January 2026, the [Chinese government announced](#) it would remove the export tax rebate for solar products on April 1, 2026.

Price of Silver (USD per gram)



Sources: BNEF, Solar Spot Index, accessed Feb. 2026; [finance.yahoo.com](#); Mercom, [In Temporary Cheer for Solar Sector, Silver Prices Crash After Months-Long Rally](#), Feb. 2026.

U.S. Distributed PV Module Prices by Origin

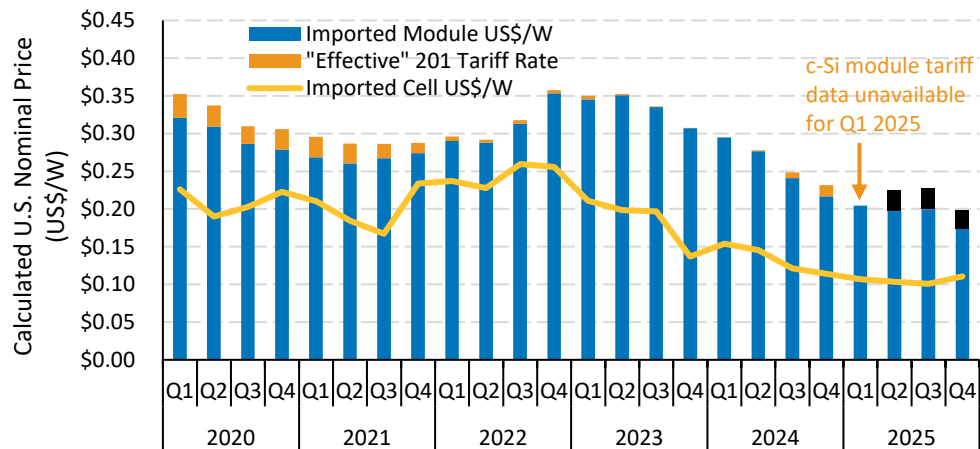


- The U.S. solar supply chain is only indirectly impacted by Chinese market conditions due to tariffs and other trade restrictions.
- Anza reported PV modules for distributed applications were relatively stable in Q4 2025, as the industry awaits a decision by the federal government on the Section 232 investigation of polysilicon and its derivatives.
 - Anza expects a price disruption once a decision is made.
- Anza reported a price premium on PV modules that have a higher degree of domestic production.
 - In 2025, the data demonstrated an increase in the price of U.S. manufactured modules using imported cells as project owners attempt to rely on non-cell module components (e.g., junction box, backsheet), as well as inverters and racking, to claim the domestic content bonus.
 - Anza reported that in 2025 mono-PERC and TOPCon modules were trading at comparable pricing; however, HJT modules received a ~\$0.12/W_{dc} premium.

Calculated U.S. Module and Cell Import Pricing

Based on the reported value and capacity of imported PV modules and cells, in Q4 2025, the average price of a U.S. module fell abruptly to \$0.17/W_{dc}, while cell prices rose slightly to \$0.11/W_{dc}.

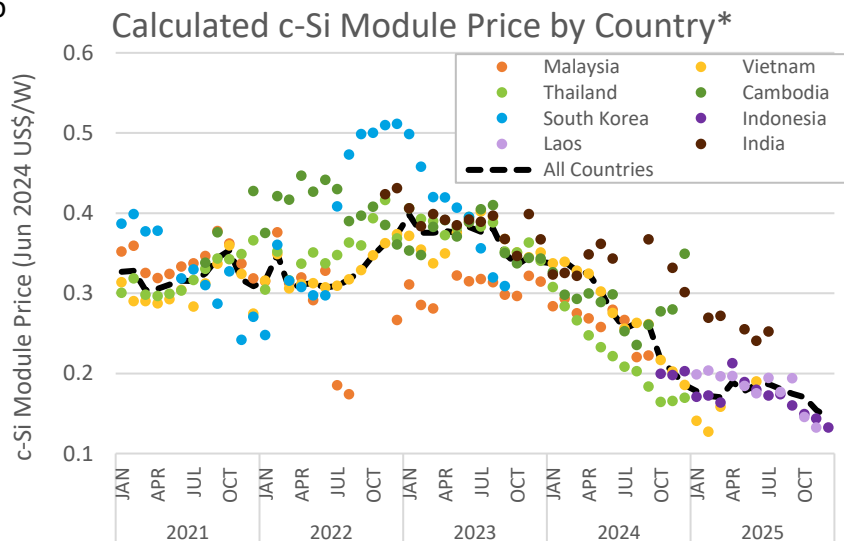
- The imposition of reciprocal tariffs in Q2 2025 rendered it impossible to measure the “effective” 201 tariff rate as all imports paid tariffs.



Note: The tariff rate was adjusted by the capacity subject to the tariffs. Manual corrections were made to three values because of suspected data entry errors for HTS code 8541430010: Cambodia (February 2022), Malaysia (June 2020), and Vietnam (July 2019). Several gigawatts of imports from India in 2022–2024 and Indonesia in 2025 entered under the HTS code for thin-film modules, but classifications may be erroneous, as the volumes exceed known manufacturing capacities for thin-film panels in India and Indonesia, based on [news reports](#). *Module price by country data plotted for select countries when import volumes >5% of total volumes.

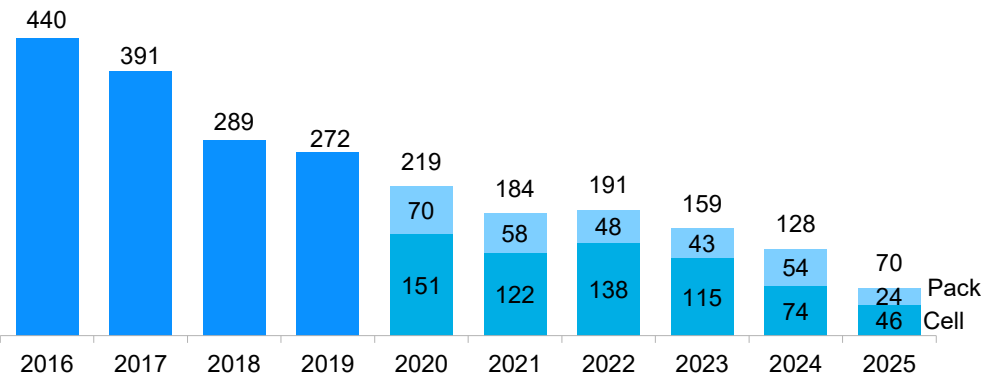
Sources: Imports by HTS code: 8541406015 (2018–2021)/8541430010 (2022–), 8541406035 (2018–2021)/8541430080 (2022–), and 8541406025 (2018–2021)/8541420010 (2022–). Second quantity (watts) from the U.S. Census Bureau [USA Trade Online Tool](#) and [Statistical Corrections](#) as of Feb. 19, 2025.

As imports from Indonesia and Laos became the dominant sources of modules in 2025, prices started at comparable prices to Vietnam and Thailand. However, starting in H2 2025, prices declined even further, ending the year at \$0.13/W_{dc} in December. Current prices remain at all-time domestic lows, when adjusted for inflation, and are approaching global values, not accounting for tariffs.



Global Average Stationary Storage Lithium-Ion Battery Pack Price, 2016–2025

Real 2025 \$ per kilowatt-hour



- After a small increase in 2022, stationary storage battery pack prices continued their historical downward trend in 2025, falling 84% from 2016 to 2025 and 45% between 2024 and 2025.
 - BloombergNEF reported stationary storage battery packs were approximately 29% lower than battery packs for passenger EVs in 2025 after being 29% higher in 2024.
- BloombergNEF cited significant competition in China, manufacturing capacity expansion, the adoption of lower-cost LFP technology, and the movement to larger cell sizes, as the drivers for the decrease in price of battery packs for stationary storage from 2024 to 2025.

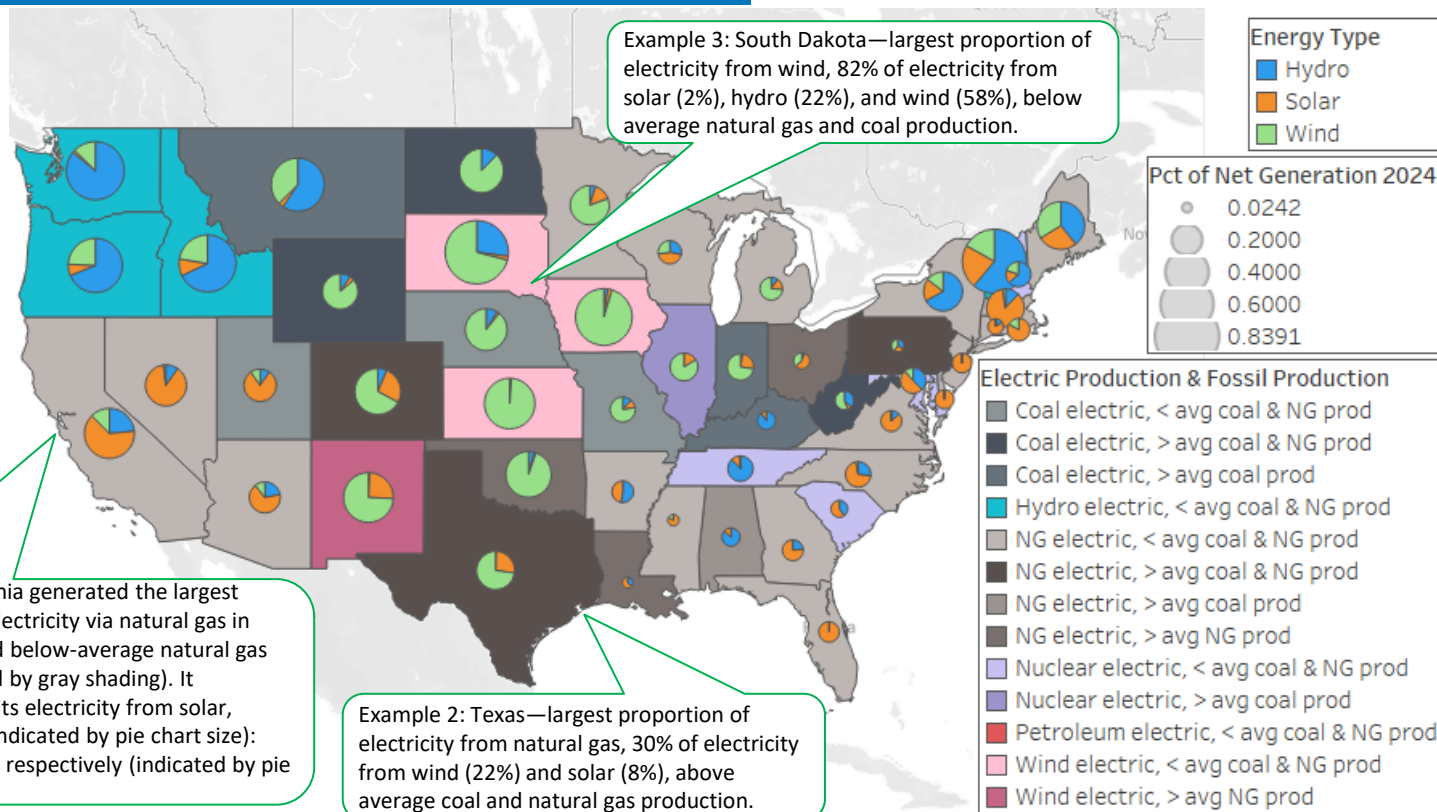
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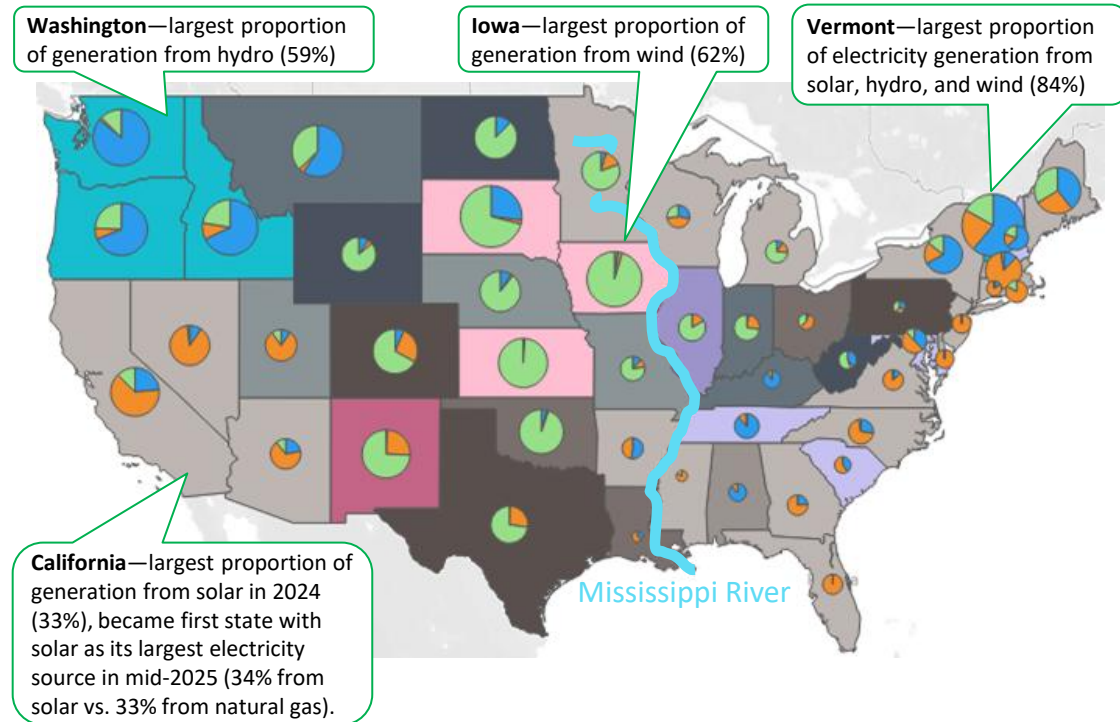
- On February 12, 2026, the IRS issued Notice 2026-15, which explains how taxpayers must determine if 45Y or 48E (tech-neutral PTC and ITC) qualified facilities and energy storage properties, or 45X eligible components (manufacturing production tax credit), include material assistance from a prohibited foreign entity (PFE).
- As of January 2026, 24 states had introduced balcony solar legislation, compared with only 3 as of fall 2025.
- In 2025, there were approximately \$35B in tax equity transactions and tax credit sales for solar, wind, and battery generating assets, and another \$10–\$15B in tax credit sales for other credits involving manufacturing, nuclear, and carbon capture.
- Top trends in 2025 changes to state distributed generation policies include transitioning to more dynamic pricing of exported electricity, fixed fees, and exploring the benefits of storage.

The State of State Electricity Generation, 2024

The state shading in the map below shows, for 2024, the largest source of electricity generation in each state along with the status of coal and natural gas production (above or below the average across states in 2024). The pie charts indicate hydro, solar, and wind generation as a proportion of net electricity generation in 2024.



The State of State Electricity Generation, 2024



In 2024, the largest proportion of electricity generation came from natural gas in 27 states, followed by coal (9 states), nuclear (5 states), hydro (4 states), wind (4 states), and petroleum (Hawaii).

On a rolling 12-month basis, solar overtook natural gas to become California's dominant electricity source in mid-2025.

Only 2 (13%) of 15 states with above-average coal or natural gas production in 2024 did not get most of their electricity from one of those resources: Illinois (nuclear) and New Mexico (wind).

- Among the 35 states with below-average coal and natural gas production, 12 (34%) got most of their electricity from another source in 2024.

In the contiguous United States, the 26 states east of the Mississippi River averaged 15% of their electricity from hydro, solar, and wind in 2024—compared with 41% for the 22 states west of the river.

- The river roughly demarcates a more sparsely populated western region with strong sun, wind, and hydro resources and an eastern region with denser population and less of those resources.

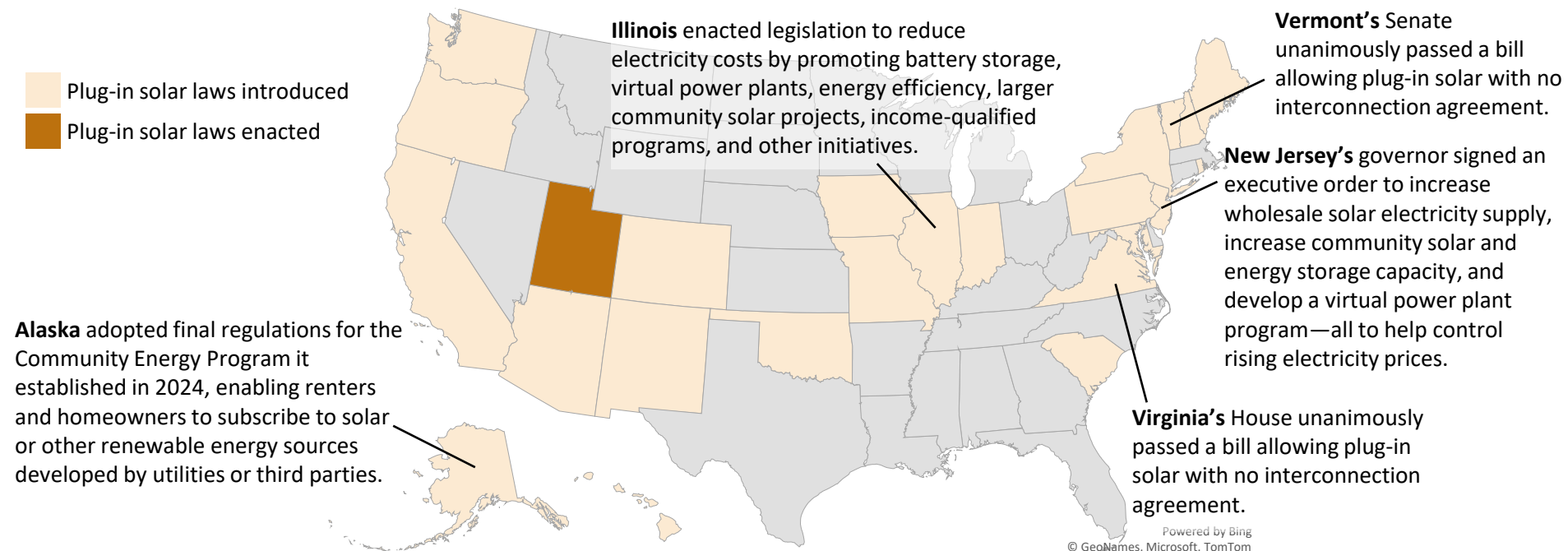
Top Distributed PV Policy Trends of 2025

- 1. Net billing transitions based on utility proposals**—e.g., regulators in West Virginia and Nevada approved net billing programs proposed by utilities in those states.
- 2. Net metering with advanced rate design structures**—e.g., Ameren Missouri approved time-of-use rates for net metering customers.
- 3. Distributed generation plus energy storage**—e.g., Illinois, Massachusetts, and New Hampshire explored the grid benefits of distributed generation plus storage.
- 4. Distributed generation cost-benefit analyses**—e.g., Massachusetts and Illinois published distributed generation cost-benefit analysis tools and avoided cost calculators.
- 5. Aggregate program cap revisions**—e.g., Kansas increased its net metering cap from 5% to 8% of peak demand, New Mexico revised its community solar cap.

- 6. Consolidated billing with community solar**—e.g., Xcel Energy included consolidated billing in its community solar proposal for Colorado.
- 7. Approval of full increases in residential fixed charges**—utilities in 11 states received a full requested increase in residential fixed charges, from \$1.00 to \$13.21 per month.
- 8. Responses to federal incentive loss**—e.g., New Jersey and Connecticut revised application periods for affected programs.
- 9. New proposed fees for distributed generation customers**—e.g., a residential peak use charge proposed by Ameren Illinois was denied, while Nevada Power's demand charges for residential and small commercial customers were approved.
- 10. Iterative revisions to net metering policies**—e.g., Maine implemented tiered credits and a new credit rate method for its net billing program.

State Policy Updates

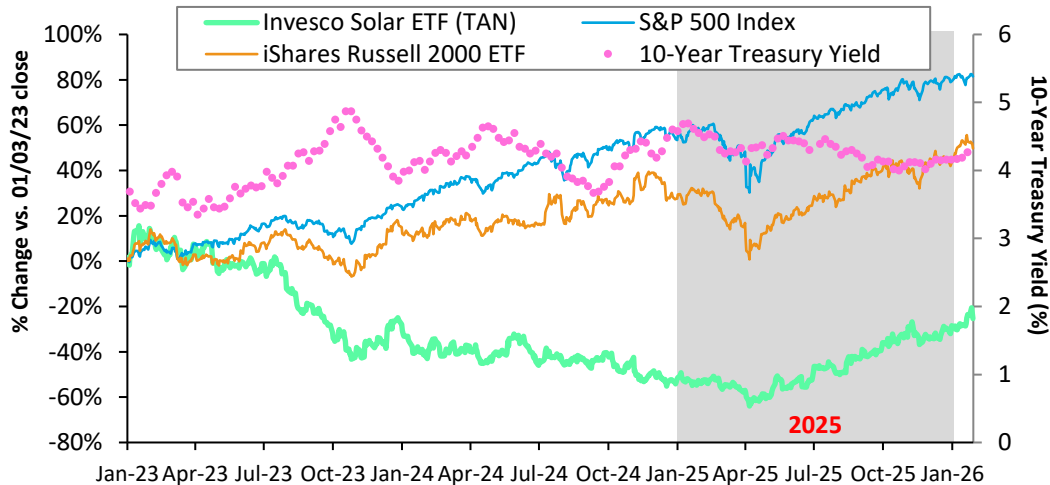
The map below shows states that have introduced or enacted legislation to allow utility customers to install plug-in solar systems without utility permission. As of January 2026, 24 states had done so, compared with only 3 as of fall 2025.



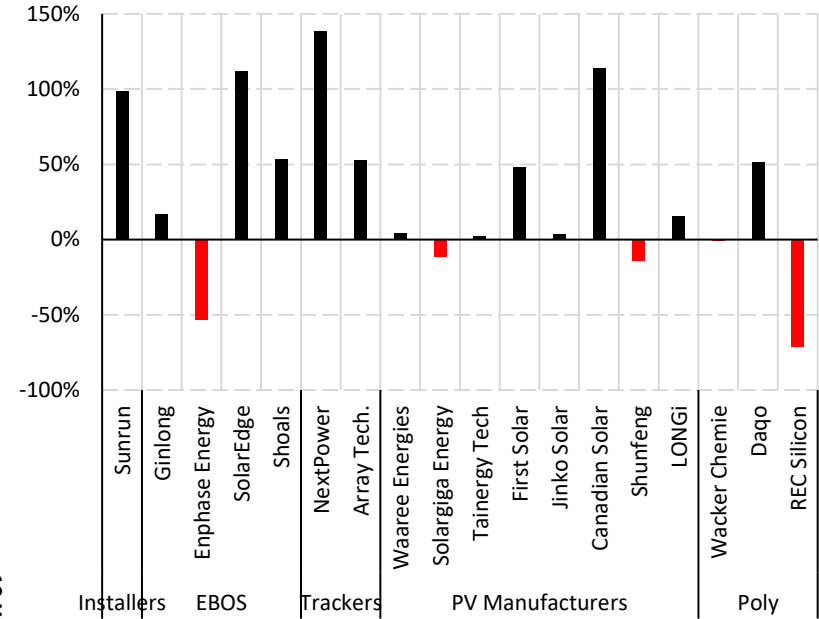
Sources: Brattleboro Reformer, [Portable Solar Looks to Plug-in](#), Feb. 2, 2026; Commonwealth of Virginia, [2026 Regular Session: HB395](#), accessed Feb. 2026; NC Clean Energy Technology Center, 50 States of Solar: Q4 2025 Quarterly Report & 2025 Annual Review, Jan. 2025; PV Magazine, [We Are Really Looking at a Movement Here](#), Jan. 30, 2026; PV Magazine, [New Jersey Governor Boosts Utility-Scale Solar, 3 GW of Community Solar, 750 MW of Storage on Day One](#), Jan. 27, 2026; Utility Dive, [Illinois Sets 3-GW Energy Storage Target, Requires Utilities To Develop Virtual Power Plants](#), Jan. 9, 2026.

Stock Market Activity

The Invesco Solar exchange-traded fund (ETF; TAN) rose 48% in 2025, compared with 11% and 16% increases for the Russell 2000 and S&P 500. Solar stock prices declined in early 2025 amid U.S. policy uncertainty but recovered midyear as it became clear how new policies would enable continued use of federal tax incentives for solar projects. Some investors also remained optimistic about solar growth because of surging electricity demand and the competitive cost of solar technologies.



Individual Stock Performance (Dec. 31, 2024–Dec. 31, 2025)



Note: The TAN index is weighted toward particular countries and sectors. As of February 2026, 50% of its funds were in U.S. companies, and 17% were in Chinese companies. Its top 10 holdings, representing 59% of its value, were NextPower, First Solar, Sunrun, Enphase, Enlight, GCL, HA Sustainable Infrastructure, Xinyi, SolarEdge, and Solaria.

Sources: Federal Reserve Bank of St. Louis, [Market Yield on U.S. Treasury Securities at 10-Year Constant Maturity](#), Feb. 2026; Invesco, [Invesco Solar ETF](#), Feb. 2026; Investing.com, [Solar Stocks Outshine S&P 500 in 2025 Despite Policy Headwinds](#), Aug. 2025; Investor's Business Daily, [The Stock Market Thought Solar Was Dead Under Trump But This Is Why Stocks Are Roaring Back](#), Aug. 2025; MarketBeat, [If Solar's Rally Has Legs, These 2 Stocks Could Benefit Most](#), Jan. 2026; MarketWatch, [Invesco Solar ETF, iShares Russell 2000 ETF, and S&P 500](#), accessed Feb. 2026; PV Magazine, [Revised Solar Stock Investment Risks](#), Aug. 2025; Solar Cell USA, [Solar Stocks Surge But Face Looming Challenges Ahead](#), May 2025.

IRS Releases Guidance on Material Assistance

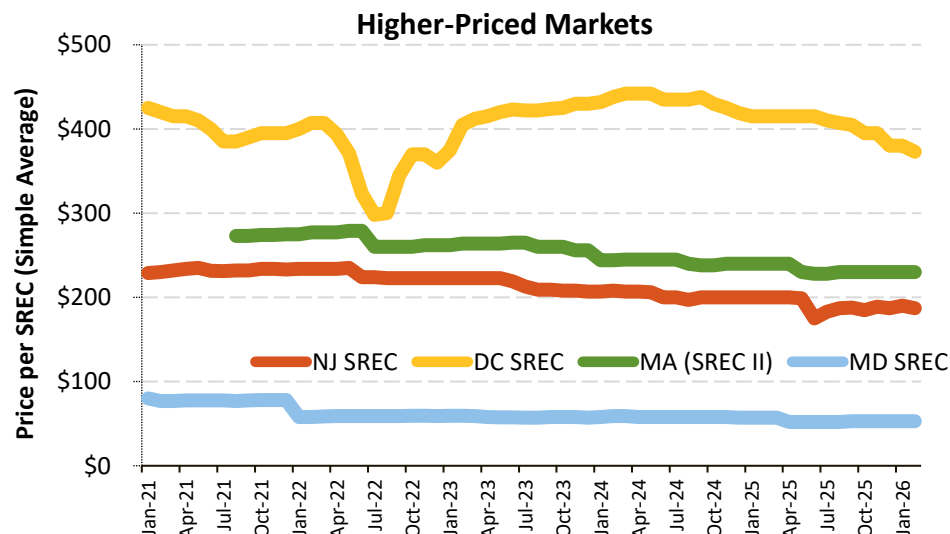
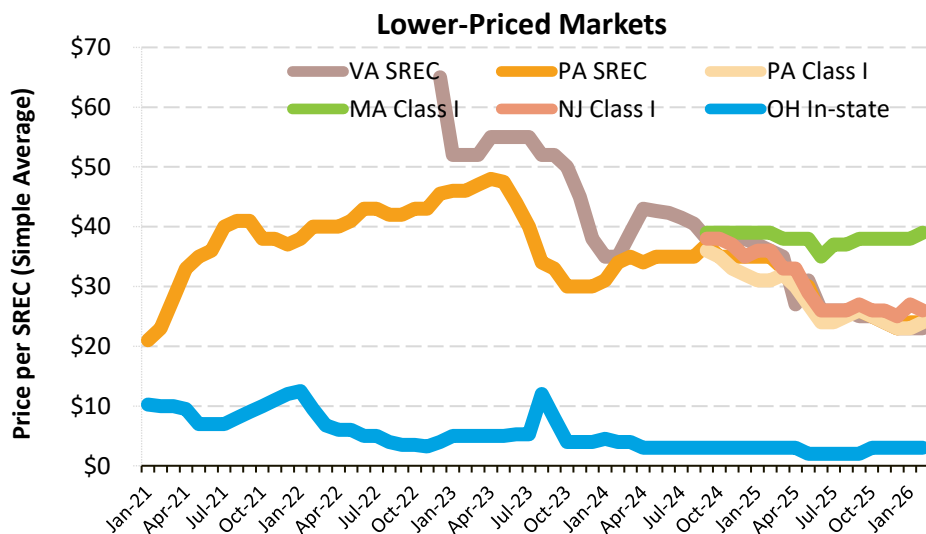
- On February 12, the IRS issued [Notice 2026-15](#), which explains how taxpayers must determine if 45Y or 48E (tech-neutral PTC and ITC) qualified facilities and energy storage properties, or 45X eligible components (manufacturing production tax credit) include material assistance from a prohibited foreign entity (PFE).
 - The PFE restrictions were enacted under 2025's HR1 and generally apply to manufactured products and projects starting in 2026. They largely restrict who can own a project and how much material can come from certain countries (e.g., China).
 - Taxpayers must calculate a project's material assistance cost ratio (MACR), which is the percentage of costs attributable to equipment, parts, and material, not supplied by a PFE. The required percentage increases over time.
- The notice details how taxpayers should calculate MACR and provides three interim safe harbor tables:
 - *Identification and cost percentage safe harbors*: Taxpayers can use the domestic content bonus safe harbor tables issued from 2023 to 2025 to (1) identify manufactured products and their components or material that should be assessed for purposes of calculating MACR and (2) use the direct cost percentages published to determine MACR. Steel items as well as any items not listed within previous safe harbor tables are not evaluated for material assistance. Tables are only available for a subset of technologies and products and not any facility under the "incremental production rule."
 - *Certification safe harbor*: Taxpayers may rely on certifications from suppliers for how much direct cost, if any, can be attributable to a PFE. The certifications must be signed under penalty of perjury, among other requirements.
- Other key elements of the notice include:
 - *Supply chain tracing*: Qualified facilities and storage properties do not need to worry about subcomponents, and 45X products only need to worry about directly purchased materials.
 - *IP licensing*: IP licenses with a PFE entered into after July 4, 2025, can result in violation of material assistance rules because PFE has "effective control" over company.
 - *80/20 rule*: Facilities qualifying under the 80/20 rule (in which used components represent 20% or less of total value) need only evaluate new products and components.
- The IRS intends to issue additional notices and new safe harbor tables and has requested comments by the end of March 2026.

Industry Reaction to IRS Releasing Guidance

- The CEO of Anza, a marketplace for solar and storage products, said the guidance was in line with expectations and provides an [“actionable pathway for project owners to qualify”](#) for the tax credit.
- The head of policy for Crux, a tax credit trading platform, also noted the increased clarity and reduced friction, relieving industry fear of further upstream tracing for compliance, but noted additional guidance was still needed on some key issues surrounding [which companies are PFEs](#).
 - The industry is still looking for more guidance on “effective control” by a PFE and [what is considered a foreign-influenced entity](#). The law firm [Morgan Lewis stated](#) the industry had hoped the guidance would confirm that IP licenses before July 4, 2025, would be grandfathered as acceptable.
 - Another law firm, [Foley Hoag](#), also noted the absence of guidance on the 10-year recapture period for tax credits pertaining to payments to specified foreign entities, as well as the “beginning of construction” requirements for PFE rules.
- The IRS sought comment on many of the issues to be addressed in future guidance.

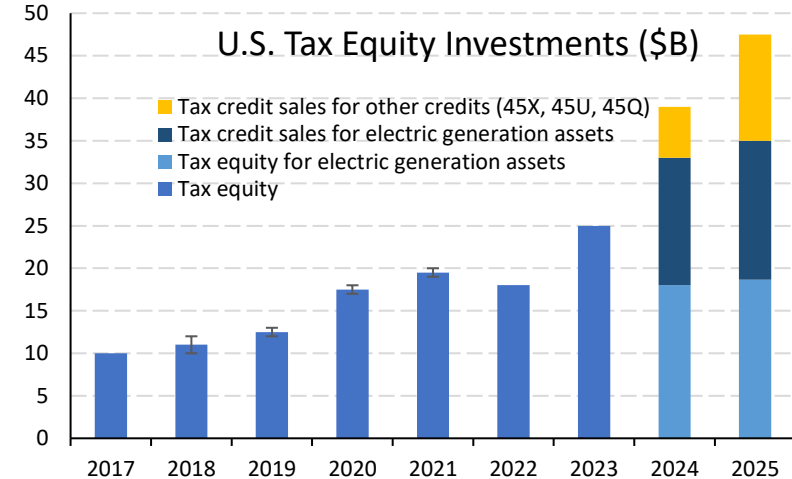
SREC and REC Pricing

- Solar renewable energy certificate (SREC) pricing was relatively stable in Q1 2026.
- Class I renewable energy certificate (REC) pricing, which is open to technologies beyond solar, tends to be significantly lower than SREC pricing, though it varies by state.



Cost of Capital, Tax Equity (Norton Rose Fulbright)

- In 2025, there were approximately ~\$35 billion tax equity transactions and tax credit sales for solar, wind, and battery generating assets, and another \$10–\$15 billion in tax credit sales for other credits involving manufacturing, nuclear, and carbon capture.
 - \$23 billion in solar and BESS, \$12 billion in wind (including repowering projects).
 - 80% of projects in marketplace are seeking to claim ITC (versus PTC).
 - Despite corporate issues within residential PV (e.g., Sunnova bankruptcy), financiers are still willing to invest in projects.
- Credits were sold in a variety of transactions, including:
 - Partnership-flip, where banks monetize tax credits
 - Straight tax credit sales
 - Hybrid partnership-flip transactions, in which some or all the credits are sold
 - Preferred equity partnership with cash investors, in which all credits are sold
 - Sale-leasebacks, which are regaining popularity.
- Financiers are arranging tax equity for projects 18 months in advance, so 2027 project deals are occurring now.
- Financiers expect a rush by lawyers, EPC firms, and financiers as solar and wind projects seek to start construction by July 4, 2026, to be eligible for tax credits after 2027.



- Currently, tax equity claiming the ITC contributes approximately 40%–50% of the total cost of a project, with projects eligible for one or more bonus credits contributing a larger portion of the total costs. However, there are more projects seeking an ITC than investors can finance, so there is a downward pressure on cost of credits (allowing investors to finance less).
 - Tax equity claiming the PTC has historically represented a larger portion of total costs; however, due to inflation and tariffs their contribution percentage is converging with projects claiming the ITC.
- To demonstrate the start of construction, most projects are building main power transformers after entering an enforceable binding contract.
 - Medium-voltage transformers, inverters, and trackers can be viewed as stock or inventory items, which would make them ineligible.

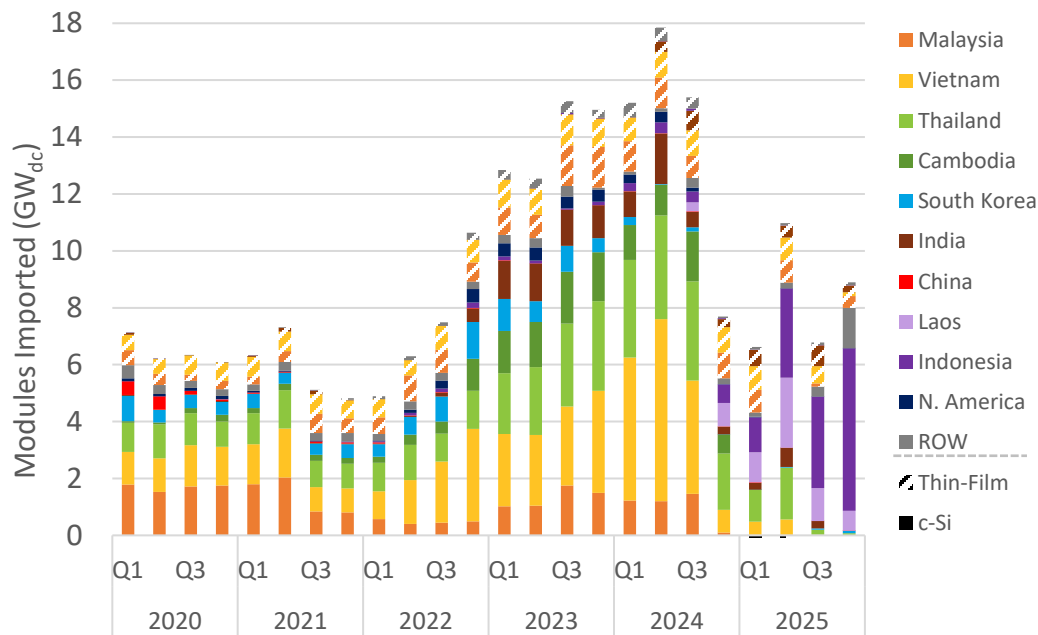
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- The United States imported 33.2 GW_{dc} of PV modules in 2025, a 40% decrease from 2024.
- The United States imported 21.1 GW_{dc} of PV cells in 2025, a 50% increase from 2024.
- Indonesia and Laos represented about half the imports of c-Si module and cell imports in 2025.

U.S. Module Imports Q4 2025 by Region

U.S. Module (c-Si + Thin-Film) Imports by Region

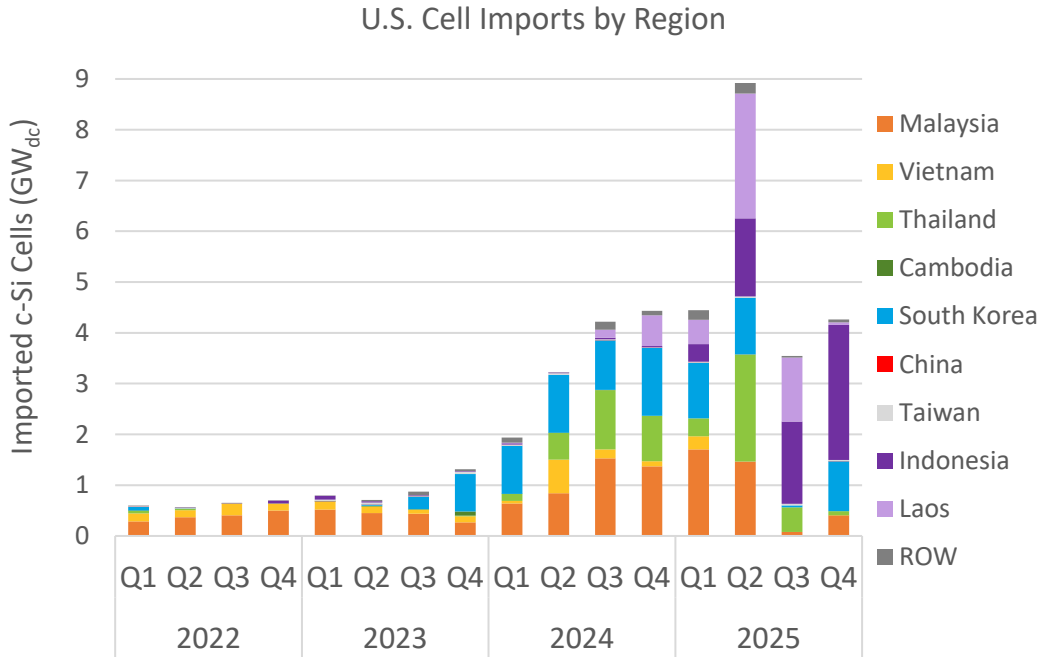


According to U.S. Census data, Q4 2025 module imports rose 31%, q/q, hitting 8.9 GW_{dc}, bringing total 2025 module imports to 33.2 GW_{dc}.

- This growth is almost entirely attributable to imports from Indonesia (64% of imports Q4 2025); imports from Thailand, Malaysia, and Vietnam have fallen to near zero.
- Imports from Ethiopia (920 MW, 10%) also grew to represent a significant fraction of imports in Q4 2025.

Note: Several gigawatts of imports from India in 2022–2024 and Indonesia in 2025 entered under the HTS code for thin-film modules, but classifications may be erroneous, as the volumes exceed known manufacturing capacities for thin-film panels in India and Indonesia. **Sources:** Imports by HTS code: 8541460015 (2018–2021)/8541430010 (2022–) and 8541460035 (2018–2021)/8541430080 (2022–); second quantity (watts) from the U.S. Census Bureau [USA Trade Online tool](#) and [corrections page](#) as of Feb. 19, 2025.

Crystalline Si PV Cell Import Data Q4 2025



According to U.S. Census data, the United States imported 4.3 GW_{dc} of PV cells in Q4 2025, a modest 20% increase, q/q, bringing total 2025 PV cell imports to 21.1 GW_{dc}.

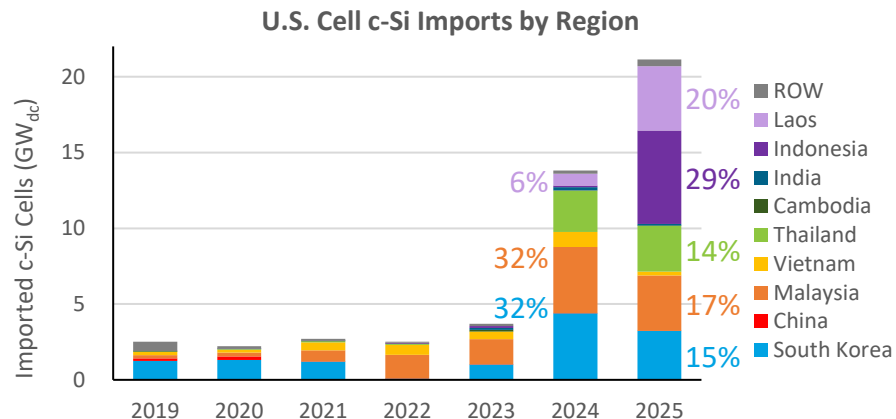
- In Q4, imports from Malaysia and South Korea restarted, likely the result of Hanwha Qcells products being temporarily detained for investigation under UFLPA. According to import data, those imports were likely released in November, with nearly 1.2 GW of cells arriving from those two countries.

On February 6, 2026, Section 201 tariffs in imported cells and modules ended.

Annual c-Si PV Cell and Module Import Data

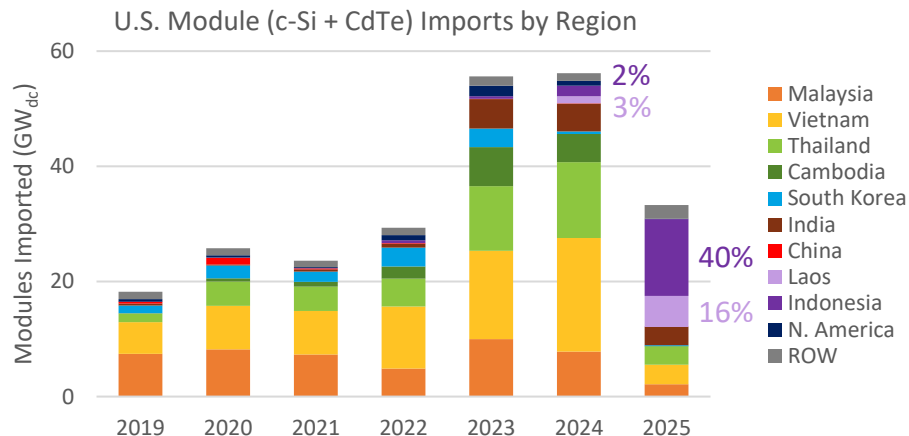
According to U.S. Census data, the United States imported 21.1 GW_{dc} of PV cells in 2025, a 50% increase from 2024.

Indonesia now represents 29% of imports, Laos 20%, up from 6% collectively in 2024.



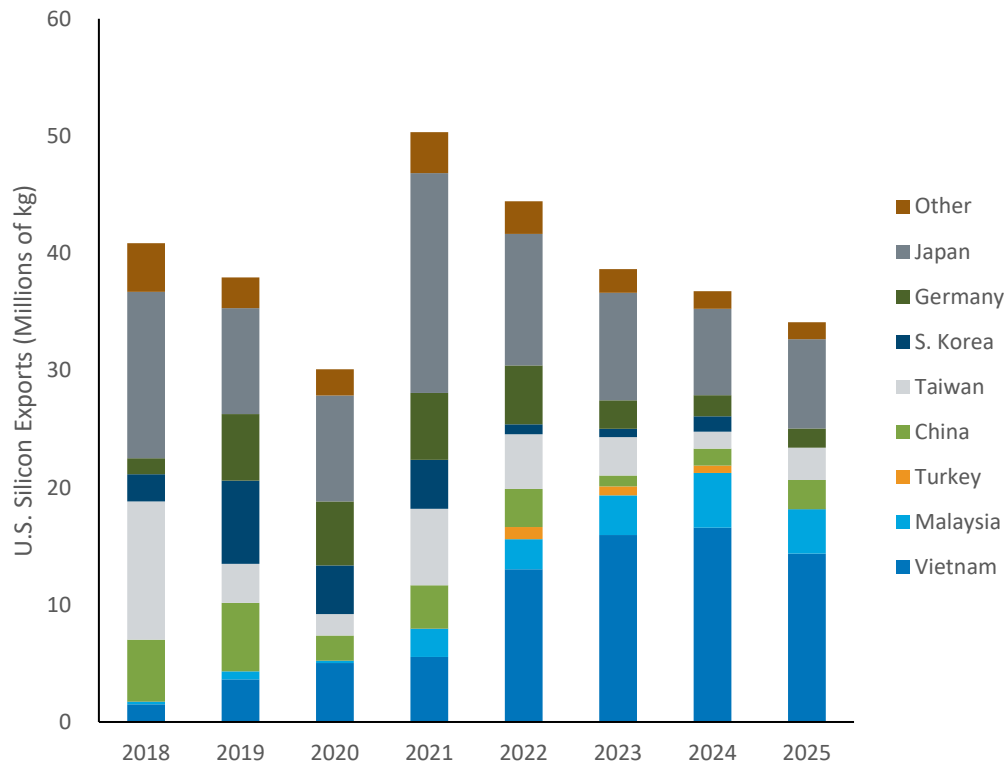
According to U.S. Census data, the United States imported 33.2 GW_{dc} of PV modules in 2025, a 40% decrease, y/y.

Similarly to cell imports, modules have shifted significantly to Laos and Indonesia, which grew from 5% of imports in 2024 to 56% in 2025.



Note: Several gigawatts of imports from India entered under the HTS code for thin-film modules in 2022–2024, but classifications may be erroneous, as the volumes exceed known manufacturing capacities for thin-film panels in India. **Sources:** Imports by HTS code: 8541460015 (2018–2021)/8541430010 (2022–), 8541460035 (2018–2021)/8541430080 (2022–), and 8541460025(2018–2021)/8541420010(2022–); second quantity (watts) from the U.S. Census Bureau [USA Trade Online Tool](#) and [corrections page](#) as of Feb. 19, 2026; Bloomberg, [US Sets Initial Duties on Indian Solar Imports at 126%](#).

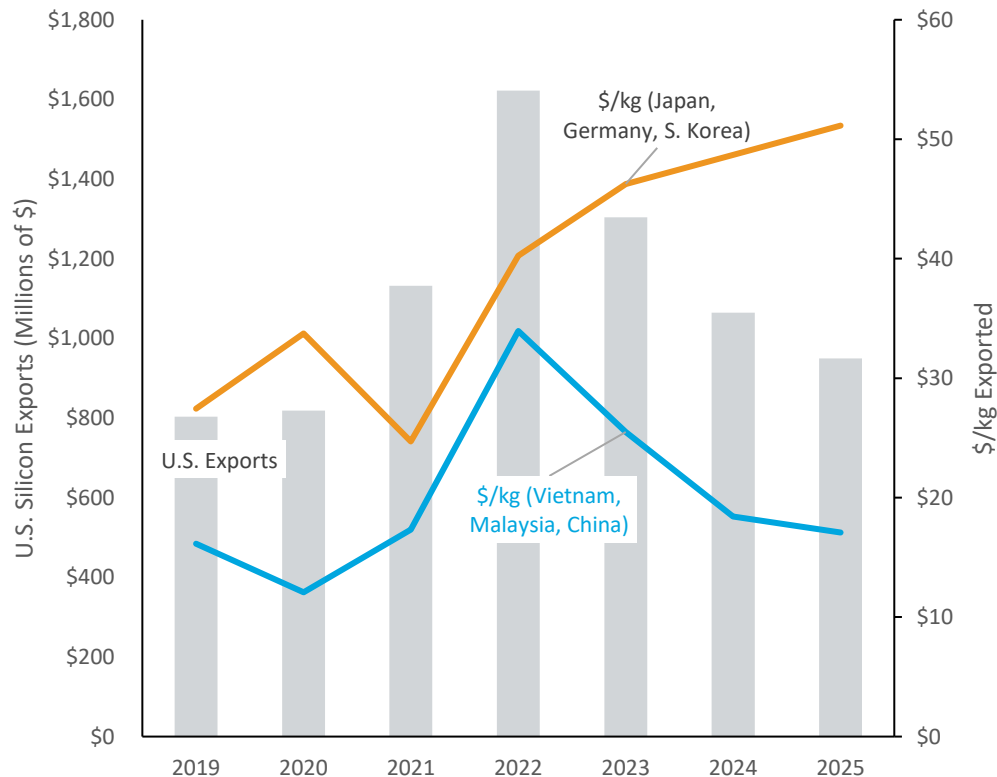
U.S. Silicon Exports (>99.99% Purity)



- After China placed duties of 57% on U.S.-manufactured polysilicon, the United States shifted exports to other markets.
 - Once China had fully scaled up its own polysilicon manufacturing industry, U.S. exports went almost exclusively to the semiconductor industry.
- Since 2018, an increasing share of U.S. exported silicon has gone to countries that manufacture a significant amount of PV wafers (e.g., Vietnam, Malaysia, Taiwan).
 - U.S. trade policies against China (e.g., UFLPA) have created a bifurcated market, resulting in more demand for U.S. polysilicon for use in PV products.
 - Since 2024, PV wafer manufacturing facilities opened in Thailand (10 GW_{dc}) and India (2 GW_{dc}); however, U.S. silicon exports did not significantly increase to those countries.

Note: The United States tracks the export of silicon with a purity exceeding 99.99% (HS Code 2804610000), which is used within the semiconductor industry (9N+) for fire-resistant material (4N+, e.g., spray coating) and for PV wafers (9N). **Source:** U.S. Census Bureau.

U.S. Silicon Exports (>99.99% Purity)



- The value/kg of silicon exports shipped to countries with significant semiconductor manufacturing capacity was significantly higher than the value/kg of silicon exports to countries with significant PV manufacturing capacity.
 - Polysilicon for semiconductor products has historically traded at a premium price over polysilicon for PV, in part because of the higher purity requirements (though that has narrowed for higher-efficiency products like TOPCon).
 - The rise in value/kg of silicon going to Vietnam, Malaysia, and China in 2022 may be due to a scarcity of UFLPA-compliant polysilicon.
- The value of U.S. silicon exports declined from 2022 to 2025 as shipments shifted more to Vietnam and Malaysia.

Note: The United States tracks the export of silicon with a purity exceeding 99.99% (HS Code 2804610000), which is used within the semiconductor industry (9N+) for fire-resistant material (4N+, e.g., spray coating) and for PV wafers (9N). **Source:** U.S. Census Bureau.



Solar Industry Update

National Lab of the Rockies | Colorado, United States

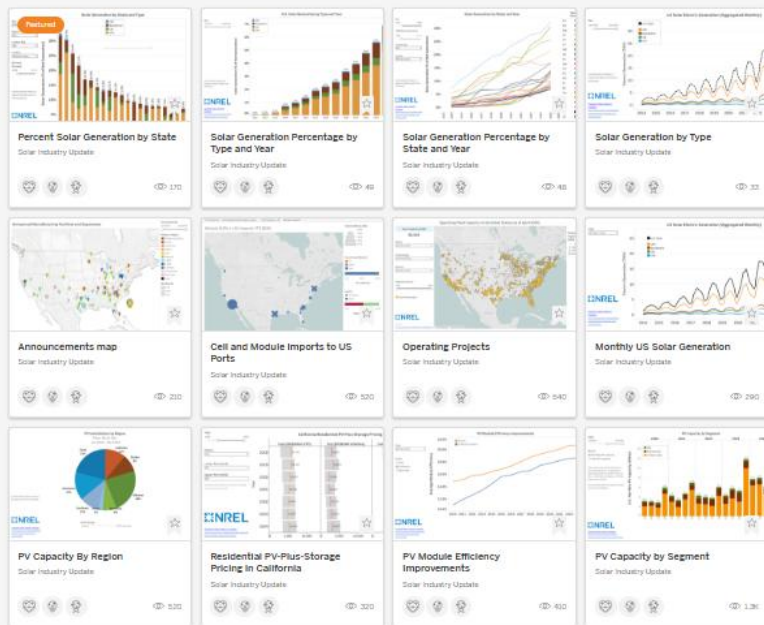
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Thank You

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List of Acronyms and Abbreviations

ac: alternating current	HTS: harmonized tariff schedule	PTC: production tax credit
B: billion	IEA: International Energy Agency	PV: photovoltaics
BESS: battery energy storage system	IP: intellectual property	Q: quarter
BNEF: BloombergNEF	IRS: Internal Revenue Service	Q1: first quarter of year
C&I: commercial and industrial	ITC: investment tax credit	q/q: quarter over quarter
c-Si: crystalline silicon	kW: kilowatt	REC: renewable energy certificate
CAISO: California Independent System Operator	kWh: kilowatt-hour	ROW: rest of the world
CdTe: cadmium telluride	M: million	SEIA: Solar Energy Industries Association
CSP: concentrating solar power	MACR: material assistance cost ratio	SPP: Southwest Power Pool
dc: direct current	MISO: Midcontinent Independent System Operator	SREC: solar renewable energy certificate
DOE: U.S. Department of Energy	MW: megawatt	STEPS: Stated Policies Scenario
DPV: distributed PV	MWh: megawatt-hour	TAN: Invesco Solar ETF
EIA: U.S. Energy Information Administration	MT: metric ton	TOPCon: tunnel oxide passivated contact
ERCOT: Electric Reliability Council of Texas	NEM: net energy metering	TW: terawatt
ETF: exchange-traded fund	NLR: National Laboratory of the Rockies	UFLPA: Uyghur Forced Labor Prevention Act
EU: European Union	NYISO: New York Independent System Operator	UPV: utility-scale PV
EV: electric vehicle	NYSERDA: New York State Energy Research & Development Authority	USD: U.S. dollars
FEOC: foreign entity of concern	O&M: operations and maintenance	USEER: United States Energy & Employment Report
GHGP: Greenhouse Gas Protocol	OBBA: One Big, Beautiful Bill Act	VRE: variable renewable energy
GW: gigawatt	PERC: passivated emitter and rear cell	W: watt
GWh: gigawatt-hour	PFE: prohibited foreign entity	WEO: World Energy Outlook
H1: first half of year	PPA: power purchase agreement	y/y: year over year
HR1: HR 1 – the One Big Beautiful Bill Act		